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10.5.10

Enterprise Data Preparation Administrator Guide

Informatica Enterprise Data Preparation Administrator Guide
10.5.10
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Preface

Use the *Enterprise Data Preparation Administrator Guide*to understand how to configure and monitor the Enterprise Data Preparation application. Learn how to set up user accounts, manage schedules, and manage user access to the data lake.

Informatica Resources

Informatica provides you with a range of product resources through the Informatica Network and other online portals. Use the resources to get the most from your Informatica products and solutions and to learn from other Informatica users and subject matter experts.

Informatica Network

The Informatica Network is the gateway to many resources, including the Informatica Knowledge Base and Informatica Global Customer Support. To enter the Informatica Network, visit https://network.informatica.com.

As an Informatica Network member, you have the following options:

- Search the Knowledge Base for product resources.
- View product availability information.
- Create and review your support cases.
- Find your local Informatica User Group Network and collaborate with your peers.

Informatica Knowledge Base

Use the Informatica Knowledge Base to find product resources such as how-to articles, best practices, video tutorials, and answers to frequently asked questions.

To search the Knowledge Base, visit https://search.informatica.com. If you have questions, comments, or ideas about the Knowledge Base, contact the Informatica Knowledge Base team at KB_Feedback@informatica.com.

Informatica Documentation

Use the Informatica Documentation Portal to explore an extensive library of documentation for current and recent product releases. To explore the Documentation Portal, visit https://docs.informatica.com.

If you have questions, comments, or ideas about the product documentation, contact the Informatica Documentation team at infa_documentation@informatica.com.

Informatica Product Availability Matrices

Product Availability Matrices (PAMs) indicate the versions of the operating systems, databases, and types of data sources and targets that a product release supports. You can browse the Informatica PAMs at https://network.informatica.com/community/informatica-network/product-availability-matrices.

Informatica Velocity

Informatica Velocity is a collection of tips and best practices developed by Informatica Professional Services and based on real-world experiences from hundreds of data management projects. Informatica Velocity represents the collective knowledge of Informatica consultants who work with organizations around the world to plan, develop, deploy, and maintain successful data management solutions.

You can find Informatica Velocity resources at http://velocity.informatica.com. If you have questions, comments, or ideas about Informatica Velocity, contact Informatica Professional Services at ips@informatica.com.

Informatica Marketplace

The Informatica Marketplace is a forum where you can find solutions that extend and enhance your Informatica implementations. Leverage any of the hundreds of solutions from Informatica developers and partners on the Marketplace to improve your productivity and speed up time to implementation on your projects. You can find the Informatica Marketplace at https://marketplace.informatica.com.

Informatica Global Customer Support

You can contact a Global Support Center through the Informatica Network or by telephone.

To find online support resources on the Informatica Network, click **Contact Support** in the Informatica Intelligent Cloud Services Help menu to go to the **Cloud Support** page. The **Cloud Support** page includes system status information and community discussions. Log in to Informatica Network and click **Need Help** to find additional resources and to contact Informatica Global Customer Support through email.

The telephone numbers for Informatica Global Customer Support are available from the Informatica web site at https://www.informatica.com/services-and-training/support-services/contact-us.html.

CHAPTER 1

Introduction to Enterprise Data Preparation Administration

This chapter includes the following topics:

- Enterprise Data Preparation Overview, 10
- Architecture and Components, 11
- Enterprise Data Preparation Users, 14
- Enterprise Data Preparation Concepts, 15

Enterprise Data Preparation Overview

With the advent of big data technologies, many organizations are adopting a new information storage model called data lake to solve data management challenges. The data lake model is being adopted for diverse use cases, such as business intelligence, analytics, regulatory compliance, and fraud detection.

A data lake is a shared repository of raw and enterprise data from a variety of sources. It is possible to store large volumes of structured and unstructured data from various enterprise systems within and outside the organization. Data in the lake can include raw and refined data, master data and transactional data, log files, and machine data.

Enterprise Data Preparation helps you derive more value from the data sources such as Hive, HDFS, Microsoft Azure data warehouse using a native connector, and Microsoft Azure SQL Database using a JDBC V2 connector to make the data available to all users in the organization.

Organizations are looking to provide ways for different kinds of users to access and work with all of the data in the enterprise, within the Hadoop data lake as well data outside the data lake. They want data analysts and data scientists to be able to use the data lake for ad-hoc self-service analytics to drive business innovation, without exposing the complexity of underlying technologies or the need for coding skills. IT and data governance staff want to monitor data related user activities in the enterprise. Without strong data management and governance foundation enabled by intelligence, data lakes can turn into data swamps.

Enterprise Data Preparation is a collaborative self-service big data discovery and preparation solution for data analysts and data scientists. It enables analysts to rapidly discover and turn raw data into insight and allows IT to ensure quality, visibility, and governance. With Enterprise Data Preparation, analysts to spend more time on analysis and less time on finding and preparing data.

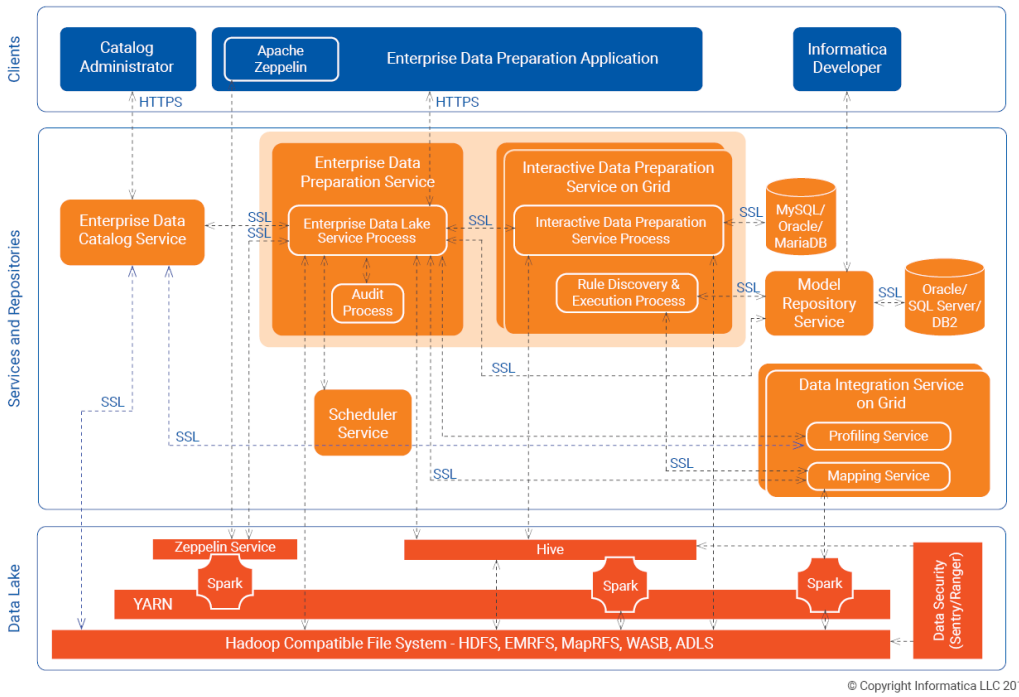
Enterprise Data Preparation provides the following benefits:

- Data analysts can quickly and easily find and explore trusted data assets within the data lake and outside the data lake using semantic search and smart recommendations.
- Data analysts can transform, cleanse, and enrich data in the data lake using an Excel-like spreadsheet interface in a self-service manner without the need for coding skills.
- Data analysts can publish data and share knowledge with the rest of the community and analyze the data using their choice of BI or analytic tools.
- IT and governance staff can monitor user activity related to data usage in the lake.
- IT can track data lineage to verify that data is coming from the right sources and going to the right targets.
- IT can enforce appropriate security and governance on the data lake
- IT can operationalize the work done by data analysts into a data delivery process that can be repeated and scheduled.

Architecture and Components

Enterprise Data Preparation uses a number of components to search, discover, and prepare data.

The following image provides an overview of the Enterprise Data Preparation deployment architecture:



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Clients

Administrators and analysts use several clients to prepare and publish data in Enterprise Data Preparation. Clients can use Hyper Text Transfer Protocol Secure (HTTPS) to connect securely to Informatica application services.

Enterprise Data Preparation uses the following clients:

Informatica Catalog Administrator

Administrators use Informatica Catalog Administrator to administer the resources, scanners, schedules, attributes, and connections that are used to create the catalog. The catalog represents an indexed inventory of all the information assets in an enterprise.

Informatica Administrator

Administrators use Informatica Administrator (the Administrator tool) to manage the application services that Enterprise Data Preparation requires. They also use the Administrator tool to administer the Informatica domain and security and to monitor the mappings run during the upload and publishing processes.

Enterprise Data Preparation application

Analysts use the Enterprise Data Preparation application to search, discover, and prepare data that resides in the data lake. Analysts combine, cleanse, transform, and structure the data to prepare the data for analysis. When analysts finish preparing the data, they publish the transformed data back to the data lake to make available to other analysts.

Informatica Developer

Administrators use Informatica Developer (the Developer tool) to view the mappings created when analysts publish prepared data in the Enterprise Data Preparation application. Administrators then operationalize the mappings so that data is regularly written to the data lake.

Apache Zeppelin

Enterprise Data Preparation integrates with Apache Zeppelin to enable users to assess and validate data by visualizing published assets.

Informatica Axon Data Governance

Enterprise Data Preparation integrates with the Axon Data Governance knowledge repository tool to provide data governance and compliance.

Application Services and Repositories

Enterprise Data Preparation requires application services and repositories to complete operations.

Use the Administrator tool to create and manage the application services and configure connections to repositories. Application services can use the Secure Sockets Layer (SSL) protocol to connect securely to other services and to repository databases.

Enterprise Data Preparation requires the following application services:

Enterprise Data Preparation Service

The Enterprise Data Preparation Service is an application service that runs the Enterprise Data Preparation application in the Informatica domain. When an analyst publishes prepared data, the Enterprise Data Preparation Service converts each recipe into a mapping.

When an analyst uploads data, the Enterprise Data Preparation Service connects to the HDFS system in the Hadoop cluster to temporarily stage the data. When an analyst previews data, the Enterprise Data Preparation Service connects to the Hadoop cluster to read from the Hive table.

Interactive Data Preparation Service

The Interactive Data Preparation Service is an application service that manages data preparation within Enterprise Data Preparation. When an analyst prepares data in a project, the Interactive Data Preparation Service connects to the Data Preparation repository to store worksheet metadata. The service connects to the Hadoop cluster to read sample data or all data from the Hive table, depending on the size of the data. The service connects to the HDFS system in the Hadoop cluster to store the sample data being prepared in the worksheet.

When you create the Enterprise Data Preparation Service, you must associate it with a Interactive Data Preparation Service.

Data Preparation repository

When an analyst prepares data in a project, the Interactive Data Preparation Service stores worksheet metadata in the Data Preparation repository.

Catalog Service

The Catalog Service is an application service that runs Enterprise Data Catalog in the Informatica domain. The Catalog Service manages the catalog of information assets in the Hadoop cluster.

When an analyst searches for assets in Enterprise Data Preparation, the Enterprise Data Preparation Service connects to the Catalog Service to return search results from the metadata stored in the catalog.

When you create the Enterprise Data Preparation Service, you must associate it with a Catalog Service.

Model Repository Service

The Model Repository Service is an application service that manages the Model repository. When an analyst creates projects, the Enterprise Data Preparation Service connects to the Model Repository Service to store the project metadata in the Model repository. When an analyst publishes prepared data, the Enterprise Data Preparation Service connects to the Model Repository Service to store the converted mappings in the Model repository.

You must associate the Enterprise Data Preparation Service with a Model Repository Service. Informatica recommends that you create a dedicated Model Repository Service and Model repository for use by Enterprise Data Preparation.

If you configure Enterprise Data Preparation to use rules, you must associate the Model Repository Service that manages the Model repository in which rule objects and metadata are stored with the Interactive Data Preparation Service.

Model repository

When an analyst creates a project, the Enterprise Data Preparation Service connects to the Model Repository Service to store the project metadata in the Model repository. When an analyst publishes prepared data, the Enterprise Data Preparation Service converts each recipe to a mapping. The Enterprise Data Preparation Service connects to the Model Repository Service to store the converted mappings in the Model repository.

Data Integration Service

The Data Integration Service is an application service that performs data integration tasks for Enterprise Data Preparation. When an analyst uploads data or publishes prepared data, the Enterprise Data Preparation Service connects to the Data Integration Service to write the data to a Hive table in the Hadoop cluster.

You must associate the Enterprise Data Preparation Service with a Data Integration Service. Informatica recommends that you create a dedicated Data Integration Service for use by Enterprise Data Preparatione.

If you configure Enterprise Data Preparation to use rules, you must associate a Data Integration Service with the Interactive Data Preparation Service. The Data Integration Service is required to run rules during data preparation.

Data Lake

The data lake used by Enterprise Data Preparation is a centralized repository of large volumes of structured and unstructured data. A data lake can contain different types of data, including raw data, refined data, master data, transactional data, log file data, and machine data.

The data lake can be deployed on-premise or in the cloud. The data lake utilizes Hive, the Spark engine, and a Hadoop-compatible file system. The data lake must be collocated with the Informatica application services associated with Enterprise Data Preparation. For example, if the data lake is deployed in Amazon EMR, you must also deploy Enterprise Data Preparation, Enterprise Data Catalog, and the Informatica services in Amazon EMR.

Enterprise Data Preparation works with the security mechanism used by the cluster, including Kerberos, Apache Ranger, and Apache Sentry, to securely access data in the data lake.

You can upload assets such as comma-separated value files as Hive tables in the data lake using Enterprise Data Preparation.

You can also ingest data from external sources outside the data lake using Hadoop tools or Informatica Mass Ingestion.

You can prepare a Hive or file-based asset that exists in Enterprise Data Catalog. When you prepare a file-based asset, Enterprise Data Preparation creates an external temporary table in the Hive schema specified in the Enterprise Data Preparation Service.

When you publish prepared data, Enterprise Data Preparation writes the transformed input source to a Hive table or to a file in the data lake.

Enterprise Data Preparation Users

Enterprise Data Preparation users include analysts who search, discover, and prepare data for analysis and administrators who manage the catalog and monitor the data lake.

Analysts work with data that reside in and outside the data lake. Analysts search for, discover, prepare, and publish data back to the data lake so that it is ready for further analysis.

Administrators use Enterprise Data Catalog to extract metadata from information assets and load the metadata into the catalog. Administrators can run Informatica mappings to load data into the data lake. They also monitor data lake usage and work with Hadoop administrators to secure the data in the data lake.

Analysts perform the following high-level tasks:

1. Search the catalog for data that resides in and outside the data lake. Discover the lineage and relationships between data that resides in different enterprise systems.
2. Create a project and prepare data.
3. Publish prepared data to the data lake to share and collaborate with other analysts.
4. Optionally use third-party business intelligence or advanced analytic tools to run reports to further analyze the published data.

Administrators perform the following high-level tasks:

1. Use Enterprise Data Catalog to create a catalog of the information assets that reside in and outside the data lake. Enterprise Data Catalog extracts metadata for each asset and indexes the assets in the catalog.
2. Operationalize the Informatica mappings created during the publication process to regularly load data with the new structure into the data lake.
   Optionally develop and run additional Informatica mappings to read data from enterprise systems and load the data to the data lake.
3. Secure the data in the data lake and monitor the usage of the data lake.

Enterprise Data Preparation Concepts

To successfully use Enterprise Data Preparation, you must understand the concepts that are used in the application.

Data Asset

A data asset is source data that you work with as a unit. A data asset can include items such as a flat file, a table, or a table view. A data asset can include data stored in or outside the data lake, such as in external databases.

You use Enterprise Data Preparation to search for assets described in the Enterprise Data Catalog, which serves as a centralized repository that stores the metadata for assets extracted from various external sources. Enterprise Data Catalog contains metadata such as profile statistics, asset ratings, data domains, and data relationships for each asset. When you search for assets in Enterprise Data Preparation, you actually search the metadata in Enterprise Data Catalog.

After you find the data asset you are interested in, you can add the data asset to a project and begin preparing the data for analysis.

Projects, Worksheets and Recipes

A project is a container that stores data assets and worksheets.

When you add a data asset to a project, Enterprise Data Preparation creates a corresponding worksheet that contains a sample of the data in a spreadsheet-like presentation. You use the worksheet to preview the data and to prepare it for publication. You do not directly change the data in the input source.

When you publish the prepared data, Enterprise Data Preparation applies the recipe to the data in the input source and creates a new data asset.

Enterprise Data Preparation captures each step you complete in the worksheet in the recipe for the worksheet. When you publish the prepared data, Enterprise Data Preparation applies the recipe to transform the data.

Data Publication

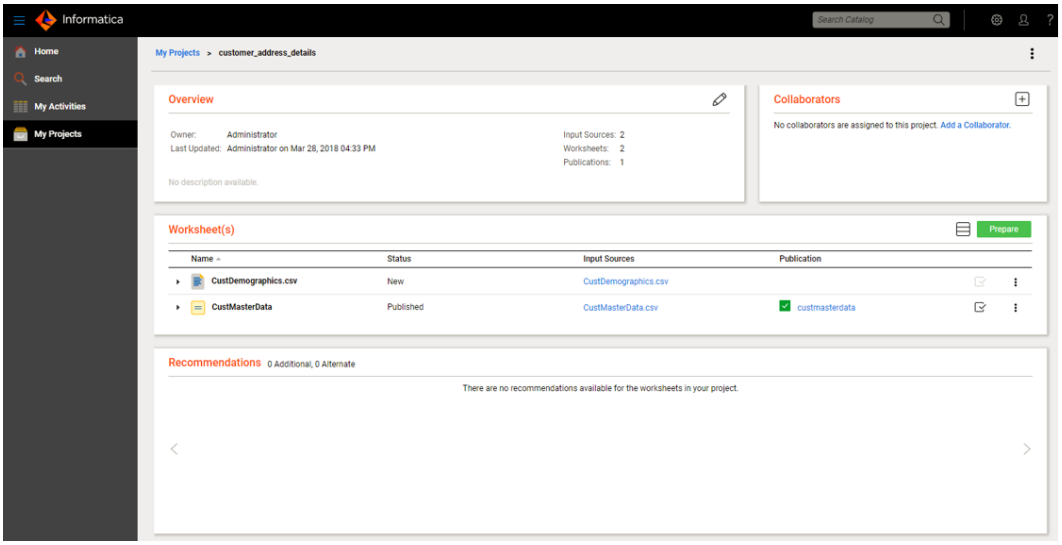
Data publication is the process of making prepared data available in the data lake.

When analysts publish prepared data, Enterprise Data Preparation writes the transformed input source to a Hive table or to a file in the data lake. Other analysts can add the published data to their projects and create

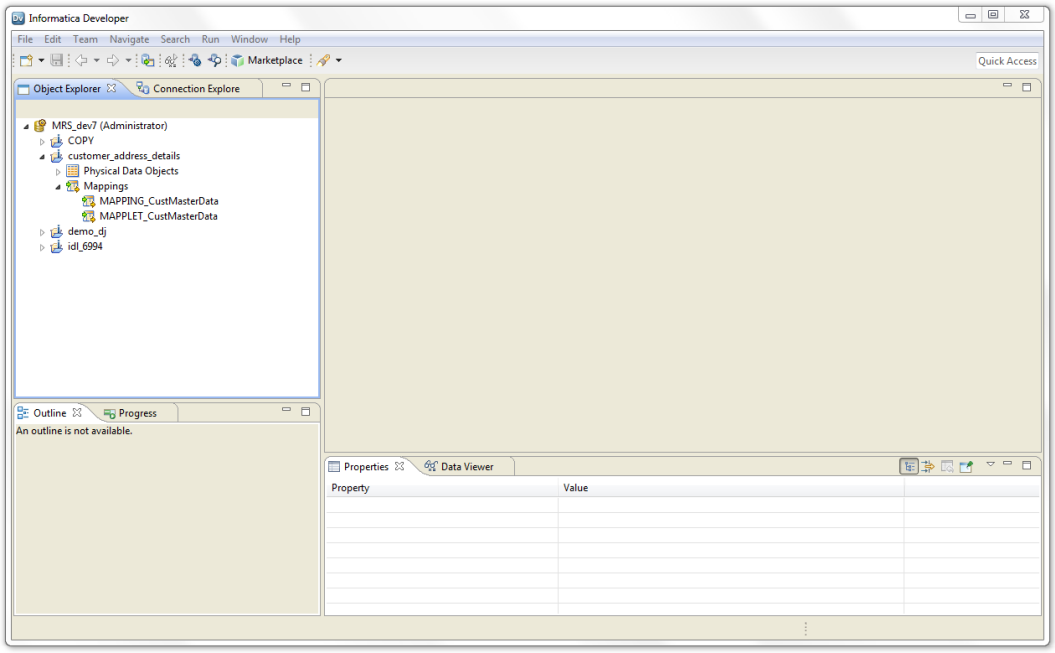
new data assets, or use a third-party business intelligence or advanced analytic tool to run reports to further analyze the published data.

Enterprise Data Preparation also converts the recipe into an Informatica mapping, and stores the mapping in the Model repository associated with the Enterprise Data Preparation Service. During the publication process, Enterprise Data Preparation uses a similar naming convention for the projects and mappings stored in the Model repository. You can use the Developer tool to view the generated mappings. You can operationalize the mappings to regularly load data with the new structure into the data lake.

The following image displays a project named `customer_address_details` in Enterprise Data Preparation. The project contains one worksheet, `CustMasterData`, with published data:



The following image displays the **Object Explorer** view in the Developer tool. The **Object Explorer** view displays the converted mapping stored in the Model repository after publication of the `CustMasterData` prepared data in the `customer_address_details` project:



Data Visualization

Data visualization is the process of assessing and validating published data using ad-hoc queries on the data to generate charts, tables, and other visual formats.

You can run queries and use dynamic forms to view the data in the form of graphs, charts, and other visual formats to assess its relevance for your business purposes. You can only run visualization on worksheets that are prepared and published. Collaborators on the project can access the notebook of the worksheet. You can share the notebook so that all users with the required permissions can view the visualization of the worksheet.

Data Preparation Process

Enterprise Data Preparation connects to several Hadoop services on a Hadoop cluster to read from and write to Hive tables and locations in the data lake, to write events, and to store sample preparation data.

Enterprise Data Preparation connects to the following services in the Hadoop cluster:

When an analyst uploads data to the data lake, the Enterprise Data Preparation Service connects to the Hadoop Distributed File System (HDFS) to stage the data in HDFS files.

When an analyst prepares data, the Interactive Data Preparation Service connects to HDFS to store the sample data being prepared in worksheets to HDFS files.

When an analyst previews data, the Enterprise Data Preparation Service connects to the Data Integration Service and reads the first 100 rows from the mapping using the JDBC driver.

When an analyst prepares data, the Interactive Data Preparation Service connects to HDFS and reads sample data from the Hive table or file, and displays the data in a worksheet.

When an analyst uploads data, the Enterprise Data Preparation Service connects to the Data Integration Service to read the temporary data staged in HDFS. If the analyst uploads data to Hive, the application writes the data to a Hive table.

When an analyst publishes prepared data, the Enterprise Data Preparation Service connects to the Data Integration Service to run the converted mappings in the Hadoop environment. The Data Integration Service applies the mapping to the data in the input source, and writes the transformed data to a Hive table in the data lake.

CHAPTER 2

Getting Started

This chapter includes the following topics:

- Getting Started Overview, 18
- Access the Administrator Tools, 18
- Configure the Enterprise Data Preparation Application Services, 20
- Configure Enterprise Data Preparation for Rules, 20
- Add Users and Assign Privileges, 22
- Create Catalog Resources, 22
- Add Catalog Resources to the Data Lake, 23
- Provide Access to Data in the Data Lake, 24

Getting Started Overview

Complete the following tasks to get started with Enterprise Data Preparation.

- Configure and enable the Enterprise Data Preparation Service and the Interactive Data Preparation Service.
- Add users and grant them privileges to access Enterprise Data Preparation.
- Create resources in Enterprise Data Catalog, and then add the resources to the data lake.
- Provide users access to the data stored in the data lake.

Access the Administrator Tools

Use Informatica Administrator and Catalog Administrator to administer the application services and the catalog that make up Enterprise Data Preparation.

Log In to Informatica Administrator

You must have a user account to log in to the Informatica Administrator web application.

If the Informatica domain runs on a network with Kerberos authentication, you must configure the browser to allow access to the Informatica web applications. In Microsoft Internet Explorer, Microsoft Edge, and Google

Chrome, add the URL of the Informatica web application to the list of trusted sites. In Safari, add the certificate of the Informatica web application to the keychain. If you are using Chrome version 86.0.42x or later on Windows, you must also set the `AuthServerWhitelist` and `AuthNegotiateDelegateWhitelist` policies.

1. Start a Microsoft Internet Explorer or Google Chrome browser.
2. In the **Address** field, enter the URL for the Administrator tool:
   - If the Administrator tool is not configured to use a secure connection, enter the following URL:

     ```
     http://<fully qualified hostname>:<http port>/administrator/
     ```
   - If the Administrator tool is configured to use a secure connection, enter the following URL:

     ```
     https://<fully qualified hostname>:<https port>/administrator/
     ```

   Host name and port in the URL represent the host name and port number of the master gateway node. If you configured secure communication for the domain, you must use HTTPS in the URL to ensure that you can access the Administrator tool.

   If you use Kerberos authentication, the network uses single sign on. You do not need to log in to the Administrator tool with a user name and password.
3. Enter the user name, password, and security domain for your user account, and then click **Login**.

   The **Security Domain** field appears when the Informatica domain contains an LDAP security domain. If you do not know the security domain that your user account belongs to, contact the Informatica domain administrator.

   Note: If this is the first time you log in with the user name and password provided by the domain administrator, change your password to maintain security.

Log In To Catalog Administrator

Use either Microsoft Internet Explorer or Google Chrome to log in to Catalog Administrator.

1. Start Microsoft Internet Explorer or Google Chrome.
2. In the Address field, enter the URL for the Catalog Administrator login page in the following format:

   ```
   http://<host>:<port>/ldmadmin
   ```

   The host is the gateway node host name. The port represents the port number configured for the Catalog Service.
3. In the Catalog Administrator login page, enter the user name and password.
4. Verify that the default domain option **Native** is selected. You can also select an LDAP domain.

   The Domain field appears when the Informatica domain contains an LDAP security domain.
5. Click **Log In**.

Configure the Enterprise Data Preparation Application Services

Enterprise Data Preparation requires the Interactive Data Preparation Service and the Enterprise Data Preparation Service. You can create the application services when you install Enterprise Data Preparation, or you can use the Administrator tool to creates the service after installation.

If you created the services during the installation, use the Administrator tool to verify that the services are correctly configured and enabled.

If you chose to create the services after the installation, use the Administrator tool to create, configure, and enable the services.

For more information on creating the Interactive Data Preparation Service, see Chapter 13, “Interactive Data Preparation Service” on page 72.

For more information on creating the Enterprise Data Preparation Service, see Chapter 14, “Enterprise Data Preparation Service” on page 86.

Tools to complete this step:

- Informatica Administrator

Configure Enterprise Data Preparation for Rules

Configure Enterprise Data Preparation to enable users to apply rules that cleanse, transform, or validate data during data preparation.

1. Use Informatica Developer (the Developer tool) to import the rule metadata into the Model repository used by the Interactive Data Preparation Service.

   For more information about importing rule metadata, see the "Importing Rules and Mappings" topic in the *Informatica Data Quality Content Installation Guide*.
2. Configure the Model Repository Service used by the Interactive Data Preparation Service for rules.
   a. Disable the Model Repository Service.
   b. Use the Administrator tool to set the following custom properties:

Custom Property	Value
CUSTOM_PROPERTIES.fined-grained-parsing	true
CUSTOM_PROPERTIES.sync-search	true

   c. Recycle the Model Repository Service.
   d. Re-index the Model Repository Service. Click **Actions**. Click **Search Index**. Click **Re-index**.
3. Verify that the Content Management Service associated with the Model Repository Service used by the Interactive Data Preparation Service is available.

4. Verify that the **Reference Data Location** property for the Content Management Service is set to the connection name for the reference data warehouse database.

   The reference data warehouse stores reference data values for the reference data objects used in rules. For example, if a rule validates U.S. zip codes, the zip code values used in the rule are stored in the reference data warehouse.

Tools to complete this step:

- Informatica Developer
- Informatica Administrator

Rule Errors

Users who use rules when preparing data might encounter errors.

The following table describes the errors that might occur when using rules:

Error Code	Description
#RULENOTFND	The rule is no longer available in the Model repository.
#RULEPERMISSIONDENIED	You do not have permissions to access the rule.
#DQMRSERR1	The application is unable to access the Model Repository Service.
#DQMRSERR2	The Model Repository Service is not available.
#DQMRSERR3	The Model Repository Service returned an error.
#DQDISSERR1	The application is unable to access the Data Integration Service.
#DQDISERR2	The Data Integration Service is not available.
#DQDISERR3	The Data Integration Service returned an error while evaluating the rule.
#AUTHERR	The user does not have permissions to connect to the Data Integration Service and the Model Repository Service.
#DQERR100	An unknown error occurred.
#LICENSE_ERR	The Informatica license is not valid for use of rules.

If a user reports an error when using rules, review the rules log file to troubleshoot the issue. The log file is available in the following directory on the node on which the Interactive Data Preparation Service runs:

```
<Informatica installation directory>/logs/<node>/services/DataPreperationService/
rule_discovery_and_execution.log
```

Add Users and Assign Privileges

Set up user accounts in the Informatica domain, and then assign the users privileges to access the Enterprise Data Preparation application. You can create user accounts in the Informatica domain or import LDAP user accounts from an LDAP directory service.

You can add the user accounts to a group to create a set of related users that have the same authorization. Assign the group the required privileges on the Enterprise Data Preparation Service or assign the group a role that includes the required privileges on the Enterprise Data Preparation Service.

Tools to complete this step:

- Informatica Administrator

Create Catalog Resources

Use Informatica Catalog Administrator to create Hive, HDFS, Microsoft Azure SQL Server, and Microsoft Azure SQL Data Warehouse resources in Enterprise Data Catalog.

A resource represents a data source from which scanners extract metadata for use in the data lake, or a data lake target to which users upload and publish data assets. Scanners attached to a resource extract metadata from the resource and store the metadata in Enterprise Data Catalog.

For more information about creating resources and scanners, see "Creating a Resource" in the *Informatica Catalog Administrator Guide*.

1. Create a Hive resource that Enterprise Data Catalog uses to extract metadata from the Hive tables in the data lake, and users access to upload or publish data to Hive. Configure the Hive resource with the following settings:
   - In the **URL** property on the **General** > **Connection Properties** panel, specify the Fully Qualified Domain Name (FQDN) of the Hive server in the JDBC connection URL.
   - If you are using operating system profiles, the Hive user name that you specify as the value for the **User** property must be a Hive superuser. For more information about operating system profiles, see "Using Operating System Profiles" on page 33.
   - Import the relevant connectors to extract metadata from Hive sources.

   For more information about Hive scanner properties, see "Hive Resource Prerequisites and Connection Properties" in the *Informatica Administrator Guide*.
2. Create an HDFS resource for each HDFS location users access to import, upload or publish data in the data lake.

   Select the **Recursive Scan** property for each HDFS resource users can access to publish data to the data lake. An error occurs if the property is not selected when a user publishes data.

   For more information about HDFS resource properties, see "HDFS Resource Connection Properties" in the *Informatica Catalog Administrator Guide*.

3. Create a Microsoft Azure SQL Server or Data Warehouse resource for each Microsoft Azure SQL Server or Data Warehouse location. Configure the resource with the following settings:
   - In the **General** > **General Properties** panel, enter the generic details and specify the resource type according to the Microsoft Azure SQL Server or Data Warehouse.
   - In the **General** > **Connection Properties** panel, specify the user name, password, host name, port number, database, and instance.
   - In the **Metadata Load Settings** panel, specify the schema and set the **Import the system objects** option.
4. Run a scan on the resources to load metadata into the catalog.
5. Create schedules for the resources so that Enterprise Data Catalog regularly scans the resources. As a best practice, schedule the resource scans to run during non-business hours.

Tools to complete this step:

- Informatica Catalog Administrator

Assign Enterprise Data Catalog Permissions

Use Catalog Administrator to set the Read and Write permission on Enterprise Data Catalog resources used by Enterprise Data Preparation. Setting the permission enables Enterprise Data Preparation users or groups of users to assign custom attributes, business terms, and data domains to assets retrieved from the catalog in search results.

Enterprise Data Preparation users must be associated with the catalog. If you add a new Enterprise Data Preparation user to the Informatica domain, you **must** run the user resource in Catalog Administrator.

For more information on assigning permissions to users and groups, see the *Catalog Administrator Guide*.

Complete the following steps to assign catalog permissions:

1. In Catalog Administrator, click **Manage** > **Security**.

   The **Security** tab page opens with the **Users and Groups** option selected in the **View** section.
2. Select **Resources** from the **View** section.

   The list of resources configured in Enterprise Data Catalog appears.
3. Select the resource for which you want to assign permissions for the users or user groups.
4. Select a user or a user group.
5. Select the **Read and Write** permission from the **Permissions** drop-down list.

Add Catalog Resources to the Data Lake

Use the Enterprise Data Preparation application to add resources created in Enterprise Data Catalog to the data lake.

You must add at least one Hive resource, which is required to perform export, import, data preparation and publish operations in the Enterprise Data Preparation application, to the data lake.

For more information about adding a resource to the data lake, see "Adding a Resource to the Data Lakehouse" on page 66.

Tools to complete this step:

- Enterprise Data Preparation application

Provide Access to Data in the Data Lake

Enterprise Data Preparation user accounts must be authorized to access the Hive tables in the Hadoop cluster designated as the data lake. Enterprise Data Preparation user accounts access Hive tables in the Hadoop cluster when they preview data, upload data, and publish prepared data.

As an administrator, grant access to the data assets that an Enterprise Data Preparation user.

When an Enterprise Data Preparation user requests access to data assets, use established processes and follow best practices to grant permission to data assets. You must grant analysts the appropriate permissions to the data lake Hadoop cluster. You can set up user impersonation to run mappings in the Hadoop cluster when an analyst publishes prepared data.

You use the tools provided with the security framework used by the Hadoop distribution to provide access to data in the data lake. For example, if you use Hortonworks with security managed by Apache Ranger, you can use the Ranger console to set up policies to grant access to Hive schemas and HDFS locations through ranger web UI. If you use Cloudera with security provided through Apache Sentry, use the Hue web application to create and manage roles and privileges.

Tools to complete this step:

- Third-party Hadoop tools

CHAPTER 3

Administration Process

This chapter includes the following topics:

- Overview of the Administration Process, 25
- Operationalize Mappings Generated from Recipes, 25
- Enable JSON Data Preparation, 26
- Monitor Data Lake Usage, 26
- Monitor the Catalog, 27

Overview of the Administration Process

Administrators use multiple tools to manage the catalog, application services, and data lake that make up Enterprise Data Preparation.

Administrators complete the following tasks to manage Enterprise Data Preparation:

- Operationalize Informatica mappings converted from recipes to regularly load data with the new structure into the data lake.
- Regularly monitor data lake usage and activity.
- Regularly monitor the catalog and the Enterprise Data Catalog tasks that extract metadata from the data lake and from enterprise systems outside the data lake.

Operationalize Mappings Generated from Recipes

You can operationalize the mappings generated by Enterprise Data Preparation to regularly load prepared data into the data lake.

When analysts publish a worksheet, Enterprise Data Preparation converts the worksheet recipe into a mapping. Enterprise Data Preparation writes the mapping to the Model repository managed by the Model Repository Service associated with the Enterprise Data Preparation Service.

Use the Developer tool to view and edit the generated mappings. The mappings use the same name as the Enterprise Data Preparation worksheet containing the published data. Verify that the mappings meet your requirements, and then use the Developer tool to deploy the mappings.

Use the Administrator tool or the infacmd command line program to run the mappings to load and transform the data. The Data Integration Service writes the data to Hive tables in the data lake. You can schedule the

deployed mappings to regularly load data into the data lake. Use the Administrator tool to monitor each mapping run.

For more information about operationalizing mappings, see "Operationalizing the Loading of Data Assets" on page 42.

Tools to complete this step:

- Informatica Developer
- Informatica Administrator

Enable JSON Data Preparation

Ensure that the following requirements are met to enable users to prepare and publish JSON files.

Hive System User Privileges

When you prepare a JSON Lines file, Enterprise Data Preparation uses the HDFS working directory configured for the Enterprise Data Preparation Service as a staging area. Enterprise Data Preparation also uses the staging schema from the Hadoop connection configured for the Enterprise Data Preparation Service.

If SQL Standards Based Authorization in HiveServer2 is enabled on Hive, ensure that the `hive` system user has read, write, and execute privileges on the directory configured as the HDFS working directory.

Cloudera CDH Configuration

If you integrate Enterprise Data Preparation with a Cloudera CDH Hadoop cluster, complete the following configuration steps to enable publication of JSON files.

1. Search for hive-hcatalog-core.jar in the Cloudera CDH installation directory.

   You can generally find the .jar file in the following directory:

   `/opt/cloudera/parcels/CDH/lib/hive-hcatalog/share/hcatalog/`
2. Log in to Cloudera Manager, and then select **Hive** in the cluster.
3. Click the **Configuration** tab, then select the **Advanced** category.
4. Enter the path to the directory containing the .jar file in the Hive Auxiliary JARs Directory property.

   If the file is in the location noted, the path to the directory is:

   `/opt/cloudera/parcels/CDH/lib/hive-hcatalog/share/hcatalog/`
5. Ensure that the following property exists in the Hive configuration:

   Property: hive.reloadable.aux.jars.path

   Value: /opt/cloudera/parcels/CDH/jars
6. Restart the Hive server after you complete the configuration updates.

Monitor Data Lake Usage

As analysts use Enterprise Data Preparation to prepare data in the data lake, you can regularly monitor the usage and activity on the data lake.

You can monitor the following activities:

Hadoop jobs that run in the Hadoop cluster when analysts upload and publish data

During the upload and publishing process, the Data Integration Service runs mappings in the Hadoop environment. The Data Integration Service pushes the processing to nodes in the Hadoop cluster. You can monitor the mappings that run during the upload and publishing processes on the **Monitor** tab of the Administrator tool. The mappings are listed as ad hoc jobs on the **Monitor** tab.

Tools to complete this step:

- Informatica Administrator
- Third-party reporting tools

Monitor the Catalog

As analysts use Enterprise Data Preparation to search for data in the catalog, you can regularly monitor the catalog and the Enterprise Data Catalog tasks that extract metadata from the data lake and from enterprise systems outside the data lake.

Use the Catalog Administrator to track the status and schedule of Enterprise Data Catalog tasks. You can monitor the duration of the tasks that are running. You can also monitor the resource distribution and track the number of resources for each resource type. To perform a detailed analysis of Enterprise Data Catalog performance, open the **Monitoring** workspace in Catalog Administrator.

Tools to complete this step:

- Informatica Catalog Administrator

CHAPTER 4

User Account Setup

This chapter includes the following topics:

- User Account Setup Overview, 28
- Step 1. Set up User Accounts, 29
- Step 2. Assign Enterprise Data Preparation Privileges, 29
- Step 3. Assign Connection Privileges, 30
- Step 4. Set Up User Access to the Data Lake, 30
- Step 5. Set Up User Impersonation, 31
- Step 6. Set Up User Access to Rules, 32
- Step 7. Set Up Access for Data Visualization, 32
- Using Operating System Profiles, 33

User Account Setup Overview

You must set up user accounts in the Informatica domain to access the Enterprise Data Preparation application and services. The Enterprise Data Preparation user accounts must also have access to the Hadoop clusters where the data lake is created.

Enterprise Data Preparation uses impersonation to provide users access to the data lake. Set up impersonation forEnterprise Data Preparation users based on the data access requirements of your organization

To set up Enterprise Data Preparation user accounts, perform the following tasks:

1. Set up user accounts in the Informatica domain.
2. Assign Enterprise Data Preparation privileges.
3. Assign connection privileges.
4. Set up user access to the data lake.
5. Set up user impersonation.

Enterprise Data Preparation uses the Data Integration Service to upload and publish data and perform operations in the data lake. The Data Integration Service uses impersonation to identify the Hadoop user that can access data and perform operations in the data lake. To use impersonation, you must configure the Data Integration Service to use operating system profiles. Follow the guidelines to use operating system profiles in Enterprise Data Preparation.

Step 1. Set up User Accounts

You set up Enterprise Data Preparation user accounts in the same manner as other user accounts in the Informatica domain. You can create Native user accounts in the Informatica domain or import LDAP user accounts from an LDAP directory service into the domain.

You can set up the following types of user accounts:

Native user accounts

You can create native user accounts for Enterprise Data Preparation. Native user accounts are created, stored, and managed in the Informatica domain.

LDAP user accounts

If you use LDAP or Kerberos authentication, you can import LDAP user accounts for Enterprise Data Preparation. Use the Administrator tool to create an LDAP security domain and set up a connection to the LDAP server. Specify the users and groups from the LDAP directory service that can have access to Enterprise Data Preparation. You can create a synchronization schedule to regularly synchronize the user accounts in the security domain and the LDAP server.

You can organize users into groups and assign privileges and data access based on the requirements of each group.

For more information about creating user accounts in the Informatica domain, see the *Informatica Security Guide*.

Step 2. Assign Enterprise Data Preparation Privileges

To enable users to log in to the Enterprise Data Preparation application, assign privileges on the Enterprise Data Preparation Service to the user accounts. If you organize users into groups, you can assign privileges to the group.

You can assign the specific privileges required by Enterprise Data Preparation or a role that has the required privileges. Informatica recommends that you assign the Administrator role to Enterprise Data Preparation users.

1. In the Administrator tool, click the **Security** tab.
2. In the Navigator, select a user or group.
3. Click the **Privileges** tab.
4. Click Edit.

   The **Edit Roles and Privileges** dialog box appears.
5. On the **Roles** tab, expand the Enterprise Data Preparation Service.

   The list of roles available for the Enterprise Data Preparation Service appears.
6. Select the Administrator role.
7. click **OK.**

Step 3. Assign Connection Privileges

Enterprise Data Preparation uses connection objects to connect to the data lake. You must assign privileges on the Hadoop connections to Enterprise Data Preparation users so they can access the data assets in the lake.

Assign Enterprise Data Preparation users privileges on the following connections to the data lake:

- Hive connection
- HDFS connection
- Cluster configuration
- JDBC connection with Sqoop

Use the same steps to assign privileges to Enterprise Data Preparation users on each connection.

1. In the Administrator tool, click **Manage** > **Connections**.
2. In the Navigator, select the corresponding data lake connection.
3. In the contents panel, select the **Permissions** view.
4. Select **Groups** or **Users**.

   You can assign permissions based on how you organize Enterprise Data Preparation users in the domain.
5. Click the **Actions** menu on the **Manage** tab.
6. Select **Assign Permission**.

   The **Assign Permissions** dialog box displays all users or groups that do not have permission on the connection. If you do not see the user or group to which you want to assign permissions, you can filter the list displayed on the dialog box by group or user name or by security domain.
7. Select a user or group, and click **Next**.
8. Select **Allow** for the permission type that you want to assign.

   Enterprise Data Preparation users require read and write permissions on each connection to the data lake.
9. Click **Finish.**

Step 4. Set Up User Access to the Data Lake

To be able to perform operations in Enterprise Data Preparation, users must have access to the data lake. Enterprise Data Preparation users must have access to the Hive data and metadata in the data lake.

Enterprise Data Preparation supports storage-based authorization. Access rights to Hive databases and tables are determined by the access rights to the underlying HDFS directories.

Assign users access to the data lake cluster based on the tasks that users perform. To preview, prepare, or export data, a user must have read access to data and metadata in the Hive database. To upload or publish data, a user must have write access to the Hive database. The user requires write access to create tables or to overwrite or append to existing tables.

The following table describes the user tasks and the permission required to perform the tasks:

Task	Permission
Data preview	Read and execute permission on the data asset.
Export	Read and execute permission on the data asset.
Data upload	Read, write, and execute permission on the temporary HDFS location. Write permission on the Hive schema where the data is uploaded.
Data preparation	Read and execute permission on the data asset to be prepared.
Publish	Read and execute permission on the data asset to be published. Write permission on the Hive schema where the data asset is published.

Step 5. Set Up User Impersonation

When a user performs tasks in Enterprise Data Preparation, the Data Integration Service connects to Hadoop services to access the data lake and perform the operation on behalf of the user. The Data Integration Service uses impersonation to pass the user account to Hadoop.

Configure user impersonation for Enterprise Data Preparation based on the user access requirements of your organization.

You can configure user impersonation in the following ways:

Using the Enterprise Data Preparation user account to connect to Hadoop services

You can configure the Data Integration Service to impersonate the user account logged in to the Enterprise Data Preparation application. The user account logged in to Enterprise Data Preparation must have user or group authorization to connect to the Hadoop services. To use this option, you must configure the Data Integration Service to use operating system profiles. You must assign the Enterprise Data Preparation user account a default operating system profile with the **Use the logged in user as Hadoop Impersonation User** option selected.

Using an authorized user account to connect to Hadoop services

If the Enterprise Data Preparation user accounts do not have logged authorization to connect Hadoop, you can configure the Data Integration Service to impersonate a specific authorized user. The impersonated user account must have authorization to connect to the data lake and the Hadoop services. To use this option, you must configure the Data Integration Service to use operating system profiles. The operating system profile must have the **Use the specified user as Hadoop Impersonation User** option selected and must specify the authorized user account to impersonate.

Using the Hive connection user account to connect to Hadoop services

If user access to the data lake does not require authorization or if all users have the same authorization, you do not need to set up operating system profiles to connect to the Hadoop services. The Data Integration Service connects to Hadoop services using the user account specified in the Hive connection object for Enterprise Data Preparation.

Configuring the User Account in the Hive Connection

The Hive connection object defines the Data Integration Service user account that can impersonate Enterprise Data Preparation users. Set the user name in the Hive connection object you use to connect to the data lake.

1. In the Administrator tool, click **Manage Connections**.
2. In the Navigator, select the Hive connection to the data lake.
3. In the properties view, edit the **Common Attribute to Both the Modes** section.

   The **Edit Common Attribute to Both the Modes** dialog box appears.
4. Set the user name and password for the Data Integration Service user account you want to use to impersonate Enterprise Data Preparation users.

   The user account must be the user account specified for user impersonation in the Hadoop cluster.
5. Click **OK.**

For more information about connection objects and the Hive connection properties, see the *Informatica Administrator Guide*.

Step 6. Set Up User Access to Rules

To be able to apply rules while preparing data in Enterprise Data Preparation, users must have access to the data quality rules.

Assign Enterprise Data Preparation users read permissions to the folder in the Model repository that contains rule objects and metadata.

You can use the Analyst tool or the Developer tool to assign users the read permission on the folder. You might want to create a group containing all Enterprise Data Preparation users, and then assign the read permission on the folder to the group.

Step 7. Set Up Access for Data Visualization

To be able to validate and assess data assets using visualization notebooks, Enterprise Data Preparation users must have access to Apache Zeppelin.

Configure users to to be able to access Zeppelin and work on it and share these details with Enterprise Data Preparation users. For more information about configuring users to access Zeppelin, see the Apache Zeppelin documentation.

If Zeppelin uses TLS authentication, the public key used by Zeppelin has to be added to the Enterprise Data Preparation truststore. For more information, see the *Administration Process* chapter.

Using Operating System Profiles

You can configure the Data Integration Service to use the permissions of the default operating system profile assigned to a user to perform operations initiated by the user.

You can assign a user a default operating system profile. If a user belongs to a group, the user inherits the default operating system profile of the group. If the user belongs to multiple groups with different default operating system profiles, the user can have multiple operating system profiles. The Data Integration Service cannot determine which profile to use for the user and cannot perform any operation initiated by the user.

Use the following guidelines when you assign an operating system profile to a user:

- A user must have only one default operating system profile.
- A user that is not assigned a default operating system profile cannot belong to multiple groups with multiple default operating system profiles.
- To avoid conflicts in group and user permissions, create a group that includes all users who require the same profile and assign a default operating system profile to the group.

For more information about creating an operating system profile, see the *Informatica Security Guide*.

For more information about configuring the Data Integration Service to use an operating system profile, see the *Informatica Application Service Guide*.

CHAPTER 5

Search Configuration

This chapter includes the following topics:

- Search Configuration Overview, 34
- Configuring Custom Attributes, 34
- Configuring System Attributes, 35
- Configuring Asset Types, 36
- Search Filter Settings, 36

Search Configuration Overview

You can configure the filter options and set the attributes that display in the search view. The data asset types and attributes that you set determine the filters that analysts can create on search results.

Configuring Custom Attributes

The Custom Attributes tab helps you configure the custom attributes that you want to display as search filters. You also add custom attributes that were recently created into the catalog. Use the configured custom attributes to refine the search.

1. Click the **Settings** icon, and then select **Search Configuration**.
2. Click the **Custom Attributes** tab.
3. Select the required custom attributes and select the following details for each attribute based on your requirements:
   - **Display in Search Results**. Displays the attribute in search results.
   - **Allow Searching**. Searches based on the attribute. You can specify the search priority for the attribute in the **Search Rank** list by selecting one of the following options:
     - **Low**
     - **Medium**
     - **High**

- **Display in Object Overview**. Displays the attribute in the different sections depending on the type of the custom attribute on the **Overview** tab.
- **Allow Sorting**. Allows you to sort the assets based on the attribute.
- **Allow Filtering**. Allows you to make the attribute a search filter on the **Filter by** panel. For custom attributes that are numeric, you can select one of the following options from the **Filter Type** list to specify how you want to specify the range of values for the filter:
  - **Slider**. displays a slider below the attribute name on the **Filter by** panel. The slider displays the lowest value configured for the attribute on the left side of the slider and the highest value on the right side of the slider. You can move the slider to the position required to select the required value for the filter. The selected value and the highest value for the filter appear in the boxes below the slider.
  - **Range**. displays boxes below the attribute name on the **Filter by** panel. You can specify the range of values in these boxes for the filter and click **GO** to perform the filtering.
- **Allow Editing**. Allows you to edit the properties of the attribute.

4. Click **Save**.

Configuring System Attributes

The System Attributes tab helps you configure the system attributes that you want to display as search filters. Use the configured system attributes to refine the search.

1. Click the **Settings** icon, and then select **Search Configuration**.
2. Click the **System Attributes** tab.
3. Select the required system attributes and select the following details for each attribute based on your requirements:
   - **Display in Search Results**. Displays the attribute in search results.
   - **Display in Object Overview**. Displays the attribute in the **System Attributes** section on the **Overview** tab.
   - **Allow Searching**. Searches based on the attribute. You can specify the search priority for the attribute in the **Search Rank** list by selecting one of the following options:
     - **Low**
     - **Medium**
     - **High**
   - **Allow Sorting**. Allows you to sort the assets based on the attribute.
   - **Allow Filtering**. Allows you to make the attribute a search filter on the **Filter by** panel.
4. Click **Save**.

   The selected system attributes appear on the **Filter by** panel.

Configuring Asset Types

The Asset Types tab helps you configure the data types that you want to display as search filters. Use the configured data types to refine the search.

1. Click the **Settings** icon, and then select **Search Configuration**.
2. Click the **Asset Types** tab.
3. Select the required asset types from the **Asset Types** section.
4. Click **Save**.

   Enterprise Data Catalog adds the selected data types to the **Asset Type** list on the **Filter by** panel.

   Note: You cannot search for the assets in the search results page if you clear to remove the assets types from the asset types list.

Search Filter Settings

The Filter Settings tab helps you customize the search filters that you view in the **Filter By** panel of the search results page. The Filter Settings tab displays the list of available filters that are expanded by default in the **Filter By** panel.

1. Click the **Settings** icon, and then select **Search Configuration**.
2. Click the **Filter Settings** tab.
3. To change the order of the search filters, select a filter and then click the **Move up** or **Move down** icon. Optionally, you can select a search filter, and then drag the filter to change the order.
4. Clear the check boxes if you want to collapse the search filters in the **Filter By** panel.
5. Click **Save**.

CHAPTER 6

Roles, Privileges, and Profiles

This chapter includes the following topics:

- Enterprise Data Preparation User Management Overview, 37
- Enterprise Data Preparation Users and Groups, 37
- Enterprise Data Preparation Service Privileges, 38
- Enterprise Data Preparation Administrator, 38
- Operating System Profiles, 39

Enterprise Data Preparation User Management Overview

You can assign privileges, roles, and permissions to users or groups of users to manage the level of access users and groups can have and the scope of the actions that users and groups can perform in the domain.

To access the application services and objects in the Informatica domain and to use the application clients, you must have a user account.

During Enterprise Data Catalog installation, a default administrator user account is created. Use the default administrator account to log in to the Informatica domain and manage application services, domain objects, and other user accounts.

Groups, privileges, and roles determine the tasks that a user can perform. You can add a user account to a group to create a set of users that have the same authorization. Assign the group or user accounts the privileges on the role that includes the required privileges on the Enterprise Data Preparation Service.

For more information about Enterprise Data Preparation user management , refer to the PDF documentation on Informatica Network. To access Informatica Network, visit https://network.informatica.com.

Enterprise Data Preparation Users and Groups

A group is a collection of users and groups that can have the same privileges, roles, and permissions.

If you add a new Enterprise Data Preparation user to the Informatica domain, you **must** run the user resource in Catalog Administrator. This is required to associate the user with the catalog and ensure that the latest changes to the catalog are available to the Enterprise Data Preparation user. For more information, see the *Catalog Administrator Guide*.

Enterprise Data Preparation Service Privileges

The Enterprise Data Preparation Service privileges determine actions that users can perform using the Enterprise Data Preparation application.

The following table lists the required permissions and the actions that users can perform with the privilege in the Data Discovery and Preparation privilege group:

Privilege Name	Permission On	Description
Data Discovery and Preparation > **Access Enterprise Data Preparation**	Enterprise Data Preparation Service	User is able to perform the following actions: - Log in to the Enterprise Data Preparation application. - Search data assets. - Upload data assets. - Prepare data assets. - Publish data assets. **Note:** You can access the application configuration details only if you are logged in as an administrator.

Enterprise Data Preparation Administrator

The Enterprise Data Preparation administrator has full permissions and privileges for managing the Enterprise Data Preparation application.

Follow these steps to create an Enterprise Data Preparation administrator:

1. In the Administrator tool, click the **Security** tab.
2. In the Navigator, select a user or group.
3. Click the **Privileges** tab.
4. Click **Edit**. The **Edit Roles and Privileges** dialog box appears.
5. On the **Roles** tab, expand the Enterprise Data Preparation Service. The list of roles available for the Enterprise Data Preparation Service appears.
6. Select the Administrator role.
7. Click **OK.**
8. Click the **Permissions** tab, and then click the **Connections** tab..
9. Assign permissions for the Hive connection, Hadoop connection, HDFS connection, anc cluster configuration created for the Enterprise Data Preparation Service.

For more information about creating users, assigning permissions, and security settings, see the *Informatica Application Security Guide*.

Operating System Profiles

An operating system profile is a type of security that the Data Integration Service uses to run mappings, workflows, and profiling jobs. Use operating system profiles to increase security and to isolate the run-time environment for users.

If the Data Integration Service runs on UNIX or Linux, create operating system profiles and configure the Data Integration Service to use operating system profiles. By default, the Data Integration Service process runs all jobs, mappings, and workflows using the permissions of the operating system user that starts Informatica services. For more information about Data Integration Service operating system profiles, see the *Informatica Application Service Guide*.

When you create the Enterprise Data Preparation Service, you must associate it with a Data Integration Service. The operating system profile configuration for the associated Data Integration Service will be applicable to the Enterprise Data Preparation Service.

Creating an Operating System Profile for Enterprise Data Preparation

Create an operating system profile to provide a level of security to users and groups in the run-time environment. The Data Integration Service you have associated with the Enterprise Data Preparation Service uses the operating system profile of the user to run workflows or jobs. Ensure that the assigned license allows you to use the operating system profiles feature.

1. In the Administrator tool, click the **Security** tab.
2. On the Security Actions menu, click **Create Operating System Profile**.

   The **Create Operating System Profile** dialog box appears.
3. Enter the following general properties for the operating system profile:

Property	Description
Name	Name of the operating system profile. The name is not case sensitive and must be unique within the domain. It cannot exceed 128 characters or begin with @. It also cannot contain the following special characters: % * + \ / . ? < > The name can contain an ASCII space character except for the first and last character. All other space characters are not allowed.
System User Name	Name of an operating system user that exists on the machines where the Data Integration Service runs. The Data Integration Service runs workflows or jobs using the system access of the system user defined for the operating system profile. **Note:** When you create operating system profiles, you cannot specify the system user name as root or use a non-root user with uid==0.

4. Click **Next**.

   The **Configure Operating System Profile** dialog box appears.
5. Select the Data Integration Service checkbox and configure the operating system profile properties.
6. Configure service process variables in the operating system profile to specify different output file locations based on the operating system profile that is assigned to the user or group. The Data Integration Service writes output files to a single shared location specified in the $DISRootDir service process variable.

7. Select **Enable Hadoop Impersonation Properties**.
8. Choose to use the logged in user or specify a Hadoop impersonation user to run Hadoop jobs. For a secure Hadoop cluster, the logged in user or the user specified as the Hadoop impersonation user should be valid and have the required permissions in the cluster nodes.
9. Optionally, configure the environment variables.
10. Click **Next**.

    The **Assign Groups and Users to Operating System Profile** dialog box appears.
11. In the Groups tab, select the groups that you want to assign the operating system profile.

    A list of all the groups with permission on the operating system profile appears.
12. In the Users tab, select the users that you want to assign the operating system profile.

    A list of all the users with permission on the operating system profile appears.
13. Click **Finish**.

    After you create the operating system profile, the details panel displays the properties of the operating system profile and the groups and users that the profile is assigned to.

Assigning a Default Operating System Profile to an Enterprise Data Preparation User or Group

In Enterprise Data Preparation, each user must have only one default operating system profile. If a user inherits multiple operating system profiles based on permissions assigned to groups or direct assignment, ensure that the user is assigned only one default operating system profile.

When you assign an operating system profile as the default profile to a user or group, the Data Integration Service uses the default operating system profile to run jobs and workflows. You can assign only an operating system profile with direct permission as the default profile to a user or group.

1. On the **Security** tab, select the **Users** or **Groups** view.
2. In the Navigator, select the user or group.
3. In the content panel, select the **Permissions** view.
4. Click the **Operating System Profiles** tab.
5. Click the **Assign or Change the Default Operating System Profile** button.

   The **Assign or Change the Default Operating System Profile** dialog box appears.
6. Select a profile from the **Default Operating System Profile** list.
7. Click **OK**.

   In the details panel, the **Default Profile** column displays **Yes (Direct)** for the operating system profile.

CHAPTER 7

Data Asset Access and Publication Management

This chapter includes the following topics:

- Data Asset Access Overview, 41
- Providing and Managing Access to Data, 41
- Data Asset Publication, 42

Data Asset Access Overview

Enterprise Data Preparation user accounts must have access to the Hive tables in the data lake when they preview data, upload data, and publish prepared data.

The administrator must follow the standards and best practices of their organization for providing data access to their users. The administrator must follow the established policies and procedures to provide the right level of access to the data asset a user requires for a project.

Providing and Managing Access to Data

Enterprise Data Preparation user accounts must be authorized to access the Hive tables on the Hadoop cluster designated as the data lake. Enterprise Data Preparation user accounts access Hive tables on the Hadoop cluster when they preview data, upload data, and publish prepared data.

HDFS permissions

Grant each user account the appropriate HDFS permissions on the Hadoop cluster. HDFS permissions determine what a user can do to files and directories stored in HDFS. To access a file or directory, a user must have permission or belong to a group that has permission to the file or directory.

A Hive database corresponds to a directory in HDFS. Each Hive table created in the database corresponds to a subdirectory. You grant Enterprise Data Preparation user accounts permission on the appropriate directory, based on whether you want to provide permission on a Hive database or on a specific Hive table in the database.

Note: As a best practice, you can set up private/shared/public databases (or schemas) in a single Hive resource and grant users appropriate permissions on those corresponding HDFS directories.

Data asset preview, import, and export

To access and preview non-lake data assets, grant users permission to connections including Oracle and Teradata Import.

For more information about Sqoop connectivity through JDBC for external databases, see the chapters on Sqoop sources and targets in the *Informatica Big Data Management User Guide*.

Hadoop user accounts

Create a user account on the Hadoop cluster for each Enterprise Data Preparation user account.

User impersonation

User impersonation allows different user accounts to run mappings in a Hadoop cluster that uses Kerberos authentication. When users upload and publish prepared data in Enterprise Data Preparation, the Data Integration Service runs mappings in the Hadoop environment. The Data Integration Service pushes the processing to nodes on the Hadoop cluster. The Data Integration Service uses the credentials you have specified to impersonate the user accounts that publish and upload the data.

When the Data Integration Service impersonates a user account to submit a mapping, the mapping can only access Hadoop resources that the impersonated user has permissions on. Without user impersonation, the Data Integration Service uses its credentials to submit a mapping to the Hadoop cluster. Restricted Hadoop resources might be accessible.

Data Asset Publication

Data publication is the process of making prepared data available in the data lake.

When analysts publish a worksheet containing prepared data, Enterprise Data Preparation applies the recipe to the data in the input source. Enterprise Data Preparation writes the transformed input source to a Hive table in the data lake.

The application also converts the recipe to a mapping, and writes the mapping to the Model repository associated with the Enterprise Data Preparation Service.

You can use a third-party business intelligence or advanced analytic tool to run reports to further analyze the published data. Other analysts can add the published data to their projects and create new data assets.

Operationalizing the Loading of Data Assets

You can operationalize the mappings created during data asset publication to regularly load data with the new structure into the data lake.

Use the Developer tool to view and edit the converted mappings stored in the Model repository associated with the Enterprise Data Preparation Service. The mappings use the same name as the Enterprise Data Preparation worksheet with the published data. Verify that the converted mappings meet your requirements, and then deploy the mappings.

Use the Administrator tool or the infacmd command line program to run the mappings to load and transform the data. The Data Integration Service writes the data to Hive tables in the data lake. You can schedule the deployed mappings to regularly load data into the data lake. Use the Administrator tool to monitor each mapping run.

For more information about running and monitoring mappings, see the *Informatica Administrator Guide*.

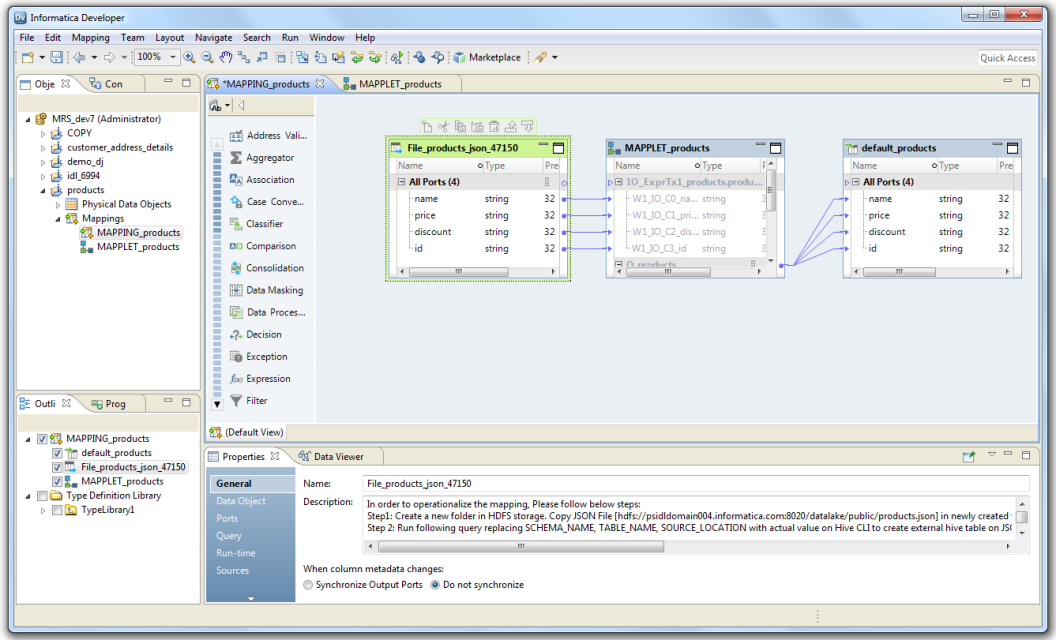
For more information about developing and running mappings that write to Hive, see the *Informatica Big Data Management User Guide*.

Operationalizing Mappings Generated for Avro, JSON, and Parquet Files

To operationalize a mapping generated for an Avro, JSON Lines, or Parquet file during publication to the data lake, you must run queries to modify Read transformations within the mapping. You must also run queries to modify Lookup transformations within a mapplet that looks up data in the file.

When you open a mapping or mapplet generated for an Avro, JSON, or Parquet file in the Developer tool, you see the syntax for two queries in the Description field for Read or Lookup transformations within the mapping or mapplet. You use the Hive CLI to run to run the first query, which creates an external table in Hive for the mapping. You then run the second query in the Developer tool to update the mapping with the external table you create in Hive.

The following image shows a mapping generated for a JSON file selected in the Developer tool. The instructions to follow to operationalize the mapping appear in the Description field in the **General** tab.



Perform the following steps on each Read and Lookup transformation within a mapping or mapplet:

1. Open a mapping or mapplet generated for an Avro, JSON, or Parquet file in the Developer tool.
2. Select a Read transformation within the mapping, or a Lookup transformation within the mapplet.
3. Click the **General** tab.
4. Copy the file from the HDFS location referenced in Step 1 to a directory in the cluster.
5. Copy the CREATE EXTERNAL TABLE query displayed for Step 2 in the Description field.
6. Replace the variables in the query with the actual values.

The following table lists the variables to set:

Variable	Description
SCHEMA_NAME	Name the Hive schema in the data lake in which to publish the data.
TABLE_NAME	Name of the external table to create.
LOCATION	Directory in the cluster to which you copied the file.

7. Use the Hive CLI to run the query on Hive.

   The query creates the external table for the file.
8. In the Developer tool, copy the SELECT query displayed for Step 3 in the Description field.
9. Click the **Query** tab.
10. Select **Advanced**, and then click **Custom Query**.
11. Paste the query into the SQL Query field.
12. Replace the variables in the query with the actual values.

    The following table lists the variables to set:

Variable	Description
SCHEMA_NAME	Name the Hive schema in the data lake in which to publish the data.
TABLE_NAME	Name of the external table created for the file.

13. Save the mapping.

    The query updates the mapping with the external table.

    If the instruction text is more than 4000 characters in length, the instruction text truncates in the Description field. If the text truncates, you can copy the queries provided in Step 2 and Step 3 directly from the publication log file.

    Note: If you publish an Avro file, you must copy the query the publication log file.

CHAPTER 8

Masking Sensitive Data

This chapter includes the following topics:

- Data Masking Overview, 45
- Configuring Data Masking, 46

Data Masking Overview

Enterprise Data Preparation integrates with Informatica Dynamic Data Masking, a data security product, to enable masking of sensitive data in data assets.

The DDM for Hive Service listens for requests from Enterprise Data Preparation. Dynamic Data Masking applies masking rules to columns in the requested table according to the masking policies defined, ensuring that data visible in the Enterprise Data Preparation application is appropriately masked for each user.

When Enterprise Data Preparation users view or perform operations on columns containing masked data, the actual data is fully or partially obfuscated based on the masking rules applied.

Data masking can be applied to data in the following scenarios:

When previewing data.

When a user previews a data asset that is in the data lake or in an external database, data is masked according to the masking rules defined on the source data.

When importing data from an external database.

When a user imports data from an external database, data masking rules are applied to the data in the Hive table created for the asset.

When preparing data.

During data preparation, columns in worksheets are populated with masked data according to the masking policies defined on the source data. When a user performs a blending operation such as a join or lookup, data in masked columns cannot be blended with unmasked columns.

When exporting or downloading data to an external database or file.

When a user exports data from the data lake to an external database, or downloads data to a file, data is written to the database or file in masked format.

When publishing data.

When a user publishes prepared data, data in columns that were masked during data preview, data sampling, and data preparation is written to the Hive table in masked format.

Configuring Data Masking

To enable data masking in Enterprise Data Preparation, you configure the Dynamic Data Masking Server to apply masking rules to data assets in the data lake. You also configure the Informatica domain to enable Enterprise Data Preparation to connect to the Dynamic Data Masking Server.

Configuring the Dynamic Data Masking Server

Use the Dynamic Data Masking Management Console to configure the Dynamic Data Masking Server.

In Dynamic Data Masking, you create the DDM for Hive service, which listens for and routes requests for the Hive data lake. You also define rules in Dynamic Data Masking to specify the masking actions to apply to columns. Dynamic Data Masking applies the data masking rules to the data before it is sent back to Enterprise Data Preparation.

For information about configuring Dynamic Data Masking, see the *Dynamic Data Masking User Guide*.

Configuring the Informatica Domain for Data Masking

Use the Administrator tool to create a Hive connection that Enterprise Data Preparation uses to connect to the Dynamic Data Masking Server. You then configure the Enterprise Data Preparation Servicee to use the connection.

Informatica recommends that you create a new Hive connection instead of modifying an existing connection, so you can revert to using the existing connection if needed.

1. Add an entry containing the IP address and host name for the Dynamic Data Masking Server host to the /etc/hosts file on each gateway node in the domain.

   Note: This step is not required if the Dynamic Data Masking Server host and Informatica domain nodes belong to the same network group.
2. In the Administrator tool, click the **Connections** tab.
3. In the Navigator, select the domain.
4. In the Navigator, click **Actions** > **New** > **Connection**.

   The **New Connection** dialog box appears.
5. In the **New Connection** dialog box, select the **Hive** connection type, and then click **OK**.

   The **New Connection** wizard appears.
6. Select the Hive connection used by Enterprise Data Preparation, and then click **Edit** in the Common Properties section.
7. Specify the properties required to connect to the Dynamic Data Masking Server.

The table below describes the properties to modify:

Property	Description
Name	The name of the connection. The name is not case sensitive and must be unique within the domain. You can change this property after you create the connection. The name cannot exceed 128 characters, contain spaces, or contain the following special characters: `~ ` ! $ % ^ & * ( ) - + = { [}] | \ : ; " ' < , > . ? /`
ID	String that the Data Integration Service uses to identify the connection. The ID is not case sensitive. It must be 255 characters or less and must be unique in the domain. You cannot change this property after you create the connection. Default value is the connection name.
Description	The description of the connection. The description cannot exceed 4000 characters.
Cluster Configuration	The cluster configuration to associate with the connection.
User Name	User name of the user that the Data Integration Service impersonates to run mappings on a Hadoop cluster. The user name depends on the JDBC connection string that you specify in the Metadata Connection String or Data Access Connection String for the native environment. If the Hadoop cluster uses Kerberos authentication, the principal name for the JDBC connection string and the user name must be the same. Otherwise, the user name depends on the behavior of the JDBC driver. With Hive JDBC driver, you can specify a user name in many ways and the user name can become a part of the JDBC URL. If the Hadoop cluster does not use Kerberos authentication, the user name depends on the behavior of the JDBC driver. If you do not specify a user name, the Hadoop cluster authenticates jobs based on the following criteria: - The Hadoop cluster does not use Kerberos authentication. It authenticates jobs based on the operating system profile user name of the machine that runs the Data Integration Service. - The Hadoop cluster uses Kerberos authentication. It authenticates jobs based on the SPN of the Data Integration Service. User Name will be ignored.
Password	Password for the user name.
Environment SQL	SQL commands to set the Hadoop environment. In native environment type, the Data Integration Service executes the environment SQL each time it creates a connection to a Hive metastore. If you use the Hive connection to run profiles on a Hadoop cluster, the Data Integration Service executes the environment SQL at the beginning of each Hive session. The following rules and guidelines apply to the usage of environment SQL in both connection modes: - Use the environment SQL to specify Hive queries. - Use the environment SQL to set the classpath for Hive user-defined functions and then use environment SQL or PreSQL to specify the Hive user-defined functions. You cannot use PreSQL in the data object properties to specify the classpath. If you use Hive user-defined functions, you must copy the .jar files to the following directory: `<Informatica installation directory>/services/shared/hadoop/<Hadoop distribution name>/extras/hive-auxjars` - You can use environment SQL to define Hadoop or Hive parameters that you want to use in the PreSQL commands or in custom queries. - If you use multiple values for the Environment SQL property, ensure that there is no space between the values.

Property	Description
SQL Identifier Character	The type of character used to identify special characters and reserved SQL keywords, such as WHERE. The Data Integration Service places the selected character around special characters and reserved SQL keywords. The Data Integration Service also uses this character for the **Support mixed-case identifiers** property.
JDBC Driver Class Name	Name of the Hive JDBC driver class.
Metadata Connection String	The JDBC connection URI Enterprise Data Preparation uses to connect to the Dynamic Data Masking Server. To connect to the data lake in Hive, specify the connection string in the following format: `jdbc:hive2://<hostname>:<port>/<schema>` Where - <hostname> is name or IP address of the machine on which the Dynamic Data Masking Server runs. - <port> is the port number on which Dynamic Data Masking Server listens. - <schema> is the name of the Hive schema in the data lake. To enable the Dynamic Data Masking Server to connect to Hive using Kerberos authentication, you must add `principal=<Hive server principal>` to the URI. For example: `jdbc:hive2://<hostname>:<port>/<schema>;principal=<Hive server principal>` For user impersonation, you must add `hive.server2.proxy.user=<xyz>` to the JDBC connection URI. If you do not configure user impersonation, the current user's credentials are used connect to the HiveServer2. If the Hadoop cluster uses SSL or TLS authentication, you must add `ssl=true` to the JDBC connection URI. For example: `jdbc:hive2://<hostname>:<port>/<db>;ssl=true` If you use self-signed certificate for SSL or TLS authentication, ensure that the certificate file is available on the client machine and the Data Integration Service machine. For more information, see the *Informatica Big Data Management Integration Guide*.
Bypass Hive JDBC Server	JDBC driver mode. Set to false to use the JDBC non-embedded mode. If you choose the non-embedded mode, you must configure the Data Access Connection String.
Fine Grained Authorization	Set to true to force mappings to observe masking rules applied to columns by Dynamic Data Masking.
Data Access Connection String	The connection string to access data from the Hadoop data store through the Dynamic Data Masking Server. Enter the connection URI you specified as the value for the Metadata Connection String property.

8. Click **Finish**.
9. In the Administrator tool, click the **Services and Nodes** tab.
10. In the Navigator, select the Enterprise Data Preparation Service.
11. Click the edit icon in the **Data Lake Options** section.
12. Select the Hive connection as the value for the **Hive Connection** property.
13. Recycle the Enterprise Data Preparation Service.

CHAPTER 9

Monitoring Enterprise Data Preparation

This chapter includes the following topics:

- Monitoring Enterprise Data Preparation Overview, 49
- Events Logged In The Audit Database , 49

Monitoring Enterprise Data Preparation Overview

Monitoring Enterprise Data Preparation involves auditing the log that stores the events and their details.

You can find answers to questions like the following by running a query on the audit database:

- Who are the top 10 most active users (in last week, last month, last year)?
- What are the top 10/50.100 recently published data assets (new data assets and updated data assets)?
- What are the largest data assets published and what are their sizes?
- What are the largest data assets uploaded and what are their sizes?
- What are the largest data assets exported and what are their sizes?
- Who exported data assets and when?
- Who created/updated/deleted/shared/viewed which project and when?
- Who added which data asset to which project and when?
- Who created/updated which recipe for a worksheet and when?

Events Logged In The Audit Database

Based on the type of activity, you can find the details of the following events saved in the audit database. You can use the event name and run a query to get the required details.

Application Access Events

LOGIN

LOGOUT

TIMEOUT

Project Management Events

- CREATE
- VIEW
- SHARE
- CHANGEOWNER
- UPDATENAME
- UPDATEDESC
- DELETE
- ADDWORKSHEET

Data Asset Related Events

- UPLOAD
- ADDDATASET
- PUBLICATION
- EXPORT
- DOWNLOAD
- IMPORT
- INGESTION
- COPY
- DELETEDATASET

Data Preparation Related Events

- PREPINEDITMODE
- PREPINVIEWMODE
- UPDATERECIPE
- RENAMEWORKSHEET
- REFRESH

Schedule Related Events

- CREATE_SCHEDULE
- DELETE_SCHEDULE
- UPDATE_SCHEDULE
- CHANGE_SCHEDULE_STATUS
- SCHEDULE_PUBLICATION

CHAPTER 10

Backing Up and Restoring Enterprise Data Preparation

This chapter includes the following topics:

- Back Up and Restore Overview, 51
- Back Up Enterprise Data Preparation, 51
- Restore Enterprise Data Preparation, 58

Back Up and Restore Overview

Informatica recommends that you periodically back up the service metatdata, repositories, and storage locations associated with Enterprise Data Preparation. Backing up Enterprise Data Preparation enables you to recover from data loss due to hardware or software failures. You should also back up Enterprise Data Preparation before you upgrade Enterprise Data Preparation.

Back up the following application services, repositories, and storage locations associated with Enterprise Data Preparation:

- Back up the Model repository associated with the Enterprise Data Preparation Service.
- Document the configuration details for the Hive scanners and HDFS scanners configured in the Enterprise Data Catalog used by the Enterprise Data Preparation Service.
- Back up the MySQL database or Oracle database used as the Data Preparation repository.
- Back up the Interactive Data Preparation Service durable storage location on HDFS.

You might also need to back up the following:

- If rules are configured for the Interactive Data Preparation Service, back up the rules metadata.

Back Up Enterprise Data Preparation

Back up the application service metatdata, repositories, and storage locations associated with Enterprise Data Preparation.

You do not need to complete the back up procedures in sequence. However, you should disable the Enterprise Data Preparation Service and the associated application services before performing a back up procedure.

Step 1. Disable the Application Services

Disable the Enterprise Data Preparation Service and all associated application services.

You use Informatica Administrator (the Administrator tool) to disable the application services. You can also use the Administrator tool to identify the application services used by the Enterprise Data Preparation Service.

Disable the application services in the following order:

- Enterprise Data Preparation Service
- Interactive Data Preparation Service
- Content Management Service
- Data Integration Service
- Model Repository Service

1. In the Administrator tool, select the **Services and Nodes** view.
2. In the Domain Navigator, select the Enterprise Data Preparation Service.
3. Click the **Disable the Service** button, and then disable the application service.
4. In the **Properties** view, click each application service associated with the Enterprise Data Preparation Service, and then disable the service.

Step 2. Back Up the Model Repository Content

Back up the content of the Model repository associated with the Enterprise Data Preparation Service. The Model repository stores project definitions and transient publication mappings associated with projects created by analysts in Enterprise Data Preparation.

Use the Administrator tool to identify the Model Repository Service associated with the Enterprise Data Preparation Service. You also use the Administrator tool to back up the Model repository. When you back up a Model repository, the Model Repository Service writes the repository contents to a backup file in the service backup directory. The service backup directory is a subdirectory of the node backup directory.

1. In the Administrator tool, select the **Services and Nodes** view.
2. In the Domain Navigator, select the Enterprise Data Preparation Service.
3. In the **Properties** view, expand **Model Repository Service Options**.
4. Select the Model Repository Service.
5. On the **Manage** tab **Actions** menu, select **Repository Contents** > **Back Up**.

   The **Back Up Repository Contents** dialog box appears.
6. Enter the options required to back up the Model Repository Service.

   The following table describes the options required to back up the Model Repository Service:

Option	Description
Username	User name of any user in the domain.
Password	Password of the domain user.
SecurityDomain	Domain to which the domain user belongs. Default is native.

Option	Description
Output File Name	Name of the backup file.
Description	Description of the contents of the backup file.

7. Click **Overwrite** to overwrite a file with the same name.
8. Click **OK**.

   The Model Repository Service writes the backup file to the service backup directory.

Step 3. Document the Configuration Properties for the Scanner Resources

Document the configuration details for each Hive scanner resource and each HDFS scanner resource configured in the Enterprise Data Catalog used by the Enterprise Data Preparation Service.

Use the Administrator tool to identify the Catalog Service associated with the Enterprise Data Preparation Service. Use the Catalog Administrator to find the configuration properties for each scanner resource.

1. In the Administrator tool, select the **Services and Nodes** view.
2. In the Domain Navigator, select the Enterprise Data Preparation Service.
3. In the **Properties** view, expand **Catalog Service Options**.
4. Select the Catalog Service.
5. Click the Catalog Administrator URL.

   The URL has the following format:

   ```
   <Catalog Service host name>:<Catalog Service port>/ldmadmin
   ```
6. In Catalog Administrator, click **Open**.

   The Library view opens.
7. Select a scanner resource.
8. Document the properties shown on each tab in Catalog Administrator.

Step 4. Back Up the Enterprise Data Preparation Connections

Back up the connection objects for the HDFS, Hive, and Hadoop connections used by the Enterprise Data Preparation Service.

Use the infacmd isp ExportDomainObjects command to export the connection object for each connection to an .xml file.

You can create an export control file to filter the objects to export from the domain. An export control file is an .xml file that you use with the infacmd isp ExportDomainObjects command. Specify the name of each connection object you want to export in an `<objectList>` element within the export control file.

The following example shows an export control file that filters the connection objects for the HDFS, Hive, and Hadoop connections used by the Enterprise Data Preparation Service:

```
<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<exportParams xmlns="http://www.informatica.com/tools/exportControl/9">
  <objectList type="connection" >
    <object name="HDFS_CCO" />
  </objectList>
```

```
  <objectList type="connection" >
    <object name="HIVE_CCO" />
  </objectList>
  <objectList type="connection" >
    <object name="HADOOP_CCO" />
  </objectList>
</exportParams>
```

If you use an export control file, use the Administrator tool to find the name of each connection used by the Enterprise Data Preparation Service. If you do not use an export control file, the infacmd isp ExportDomainObjects command exports all native users, native groups, roles, connections, and cluster configurations from the Informatica domain.

For more information about creating export control files and using the infacmd isp ExportDomainObjects command, see the *Informatica Command Reference*.

1. In the Administrator tool, select the **Services and Nodes** view.
2. In the Domain Navigator, select the Enterprise Data Preparation Service.
3. In the **Properties** view, expand **Data Lake Options**.
4. Copy the values for the **HDFS Connection**, **Hive Connection** and **Hadoop Connection** properties.
5. Expand **Event Logging Options**.
6. Create an export control file to filter the objects that the infacmd isp ExportDomainObjects command exports.

   Specify the name of each connection object you want to export in an `<objectList>` element within the file.
7. Run the following command:

   ```
   infacmd isp ExportDomainObjects -dn <domain name> -un <user name> -pd <password>
   -sdn <security domain> -fp <export file> -cp <export control file> -rp <true|false>
   ```

   The following table describes the infacmd isp ExportDomainObjects command options and arguments you might specify to export connection objects for the HDFS, Hive, and Hadoop connections:

Option	Argument	Description
-DomainName -dn	domain_name	Required. Name of the Informatica domain.
-UserName -un	user_name	Required if the domain uses Native or LDAP authentication. User name to connect to the domain. Optional if the domain uses Kerberos authentication. To run the command with single sign-on, do not set the user name. If you set the user name, the command runs without single sign-on.
-Password -pd	password	Required if you specify the user name.
-SecurityDomain -sdn	security_domain	Required if the domain uses LDAP authentication. Optional if the domain uses native authentication or Kerberos authentication. Name of the security domain to which the domain user belongs. If the domain uses native or LDAP authentication, the default is Native. If the domain uses Kerberos authentication, the default is the LDAP security domain created during installation.

Option	Argument	Description
-ExportFile -fp	export_file_name	Required. Path and file name of the export file. If you do not specify the file path, infacmd creates the file in the directory where you run infacmd.
-ExportControlFile -cp	export_control_file	Optional. Name and path of the export control file that filters the objects to export.
-RetainPassword -rp	true|false	Optional. Set to true to retain encrypted passwords for users and connections in the exported file. When set to false, user and connection passwords are exported as empty strings. Default is false.

The following example exports the connection objects for the HDFS, Hive, and Hadoop connections defined in the specified export control file:

```
infacmd isp ExportDomainObjects -dn InfaDomain -un Administrator -pd password -fp
"/backup/idl_connections.xml" -cp "/backup/idl_exportControl.xml" -rp true
```

Step 5. Back Up the Data Preparation Repository

Back up the MySQL database or the Oracle database used as the Data Preparation repository.

Use the Administrator tool to find the database type and schema name values configured for the repository database. Follow the instructions in the MySQL documentation or the Oracle documentation to back up the repository database.

1. In the Administrator tool, select the **Services and Nodes** view.
2. In the Domain Navigator, select the Interactive Data Preparation Service.
3. In the **Properties** view, expand **Data Preparation Repository Options**.
4. Find the values for the **Database Type** and **Schema Names** properties configured for the repository database.
5. Back up the Data Preparation repository database.

   For more information, see the MySQL documentation or the Oracle documentation.

Step 6. Back Up the Interactive Data Preparation Service Durable Storage

Back up the HDFS location where the Interactive Data Preparation Service stores data.

Use the Administrator tool to find the durable storage configuration properties. You can use the Hadoop DistCp tool to export the storage metadata to a backup location.

1. In the Administrator tool, select the **Services and Nodes** view.
2. In the Domain Navigator, select the Interactive Data Preparation Service.
3. In the **Properties** view, expand **Data Preparation Storage Options**, and then document the durable storage configuration properties.
4. On the HDFS cluster, run the following command:

   ```
   hadoop distcp hdfs://<durable storage location> <local file system location>
   ```

The following table describes the DistCp command arguments:

Argument	Description
Durable Storage Location	The HDFS storage location to back up. Specify the value of the HDFS Storage Location property displayed in the Administrator tool.
Backup Location	The path to the backup directory to which to write the storage metadata.

Step 7. Back Up the Rules Metadata

If rules are configured for the Interactive Data Preparation Service, back up the rules metadata stored in the Model repository that contains rule objects and metadata.

The rules metadata is stored in projects in the Model repository. Use the Administrator tool to find the Model Repository Service used by the Interactive Data Preparation Service. Use the infacmd tools ExportObjects command to export the rule objects, the dictionary content, and the NER and Classifier model files from a project in the Model repository.

The infacmd tools ExportObjects command creates an .xml file that contains the rule objects. The command also creates a .zip file containing the dictionary content as .dic extension flat files and the NER and Classifier models as .ner and .classifer files.

Run the infacmd tools ExportObjects command to export rules metadata from each project that contains rules metadata. When you run the command, you specify the name of a project that contains the rule objects. You also specify the name of the Model Repository Service that manages the Model repository that contains the project. The Model Repository Service must be running when you run the command.

You can create an export control file to filter the objects to export from the Model repository. An export control file is an .xml file that you use with the infacmd tools ExportObjects command. Specify the name of each rule object you want to export in an `<objectList>` element within the export control file.

The following example shows an export control file that filters the rule objects to export from the Model repository:

```
<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<exportParams xmlns="http://www.informatica.com/tools/exportControl/9">
  <folders>
    <folder recursive="false" select="all">
      <objectList type="Rule" >
        <object name="validate_address" />
      </objectList>
      <objectList type="Rule" >
        <object name="validate_USA_zip_code" />
      </objectList>
      <objectList type="Rule" >
        <object name="validate_Canadian_postal_code" />
      </objectList>
    </folder>
  </folders>
</exportParams>
```

If you do not use an export control file, the infacmd tools ExportObjects command exports all objects in the specified project.

For more information about creating export control files and using the infacmd tools ExportObjects command, see the *Informatica Command Reference*.

1. In the Administrator tool, select the **Services and Nodes** view.
2. In the Domain Navigator, select the Enterprise Data Preparation Service.

3. In the **Properties** view, expand Interactive Data Preparation Service **Properties**.
4. Select the Interactive Data Preparation Service shown.
5. In the **Properties** view, expand **Model Repository Service Properties** to identify the Model Repository Service used by the Interactive Data Preparation Service.

   You must identify each project within the Model repository that contains rules metadata.
6. If the Model Repository Service is not running, recycle the Model Repository Service.

   For more information, see "Step 6. Recycle the Application Services" on page 62.
7. Run the following command:

```
infacmd tools ExportObjects -dn <domain name> -un <user name> -pd <password>
-sdn <security domain> -pn <project name> -rs <Model Repository Service name>
-fp <export file> -ow <true|false> -cp <export control file> -oo <options>
```

   The following table describes infacmd tools ExportObjects options and arguments you might specify to back up the Content Management Service metadata:

Option	Argument	Description
-DomainName -dn	domain_name	Name of the Informatica domain.
-UserName -un	user_name	User name to connect to the domain. Optional if the domain uses Kerberos authentication.
-Password -pd	password	Password for the user name.
-SecurityDomain -sdn	security_domain	Name of the security domain to which the domain user belongs.
-ProjectName -pn	project_name	Name of the project in the Model repository from which to export the rules metadata.
-RepositoryService -rs	service_name	Model Repository Service name.
-ExportFilePath -fp	export_file_path	Path and XML file name of the export file that contains the rules metadata. You can specify an absolute path or a relative path to the file name.
-OverwriteExportFile -ow	true|false	Set to true to overwrite an existing export file. Default is false.

Option	Argument	Description
-ControlFilePath -cp	control_file_path	Path and file name of the export control file that filters the exported objects.
-OtherOptions -oo	options	Additional options to export the dictionary content and the NER and Classifier models to a zip file. Enter options using the following format: `rtm:<option_name>=<value>,<option_name>=<value>` Option names include: - codePage. Code page of the reference data. - refDataFile. Path and file name of the zip file where you want to export the reference table data.

The following example exports the rules objects defined in the specified export control file from a project named TEST_PROJECT to the dq_rules_metadata.xml file. The example also exports the dictionary content and the NER and Classifier models from the project named TEST_PROJECT to a dq_rules_artifacts.zip file.

```
infacmd tools exportObjects -dn InfaDomain -un Administrator -pd password -sdn
native
-pn TEST_PROJECT -rs MRS_dev2 -fp "/export/dq_rules_metadata.xml" -ow true
-cp "/dq_rules_export.xml" -oo "rtm:codePage=UTF-8,refDataFile="/output/
dq_rules_artifacts.zip""
```

Restore Enterprise Data Preparation

Restore the application service metatdata, repositories, and storage locations associated with Enterprise Data Preparation.

You do not need to complete the restore procedures in sequence. However, you should disable the Enterprise Data Preparation Service and the associated application services before performing a restore procedure.

Step 1. Restore the Model Repository Content

Restore the Model repository content from the repository backup file.

Verify that the Model repository is empty. If the Model repository contains content, the restore option is disabled. You must delete the content from the Model repository before restoring the repository content from the backup file.

1. In the Administrator tool, select the select the **Services and Nodes** view.
2. In the Domain Navigator, select the Model Repository Service associated with the Enterprise Data Preparation Service.
3. If the Model repository contains content, delete the content. From the **Actions** menu, select **Repository Contents** > **Delete**.
4. From the **Actions** menu, select **Repository Contents** > **Restore**.

   The **Restore Repository Contents** dialog box appears.
5. Select the Model repository backup file to restore.

6. Enter the properties required to restore the Model repository.

   The following table describes the properties:

Property	Description
Username	User name of any user in the domain.
Password	Password of the domain user.
Security Domain	Security domain to which the domain user belongs. Default is native.

7. Click **OK**.

Step 2. Create the Hive and HDFS Resources in the Catalog

Create the Hive and HDFS scanner resources in the Enterprise Data Catalog used by the Enterprise Data Preparation Service.

For each scanner resource, enter the property values you copied during the backup phase into each tab in the Resource view in Catalog Administrator.

For more information about creating resources, see the *Informatica Catalog Administrator Guide*.

1. In the Administrator tool, select the **Services and Nodes** view.
2. In the Domain Navigator, select the Enterprise Data Preparation Service.
3. In the **Properties** view, expand **Catalog Service Options**.
4. Select the Catalog Service.
5. Click the Catalog Administrator URL.

   The URL has the following format:

   ```
   <Catalog Service host name>:<Catalog Service port>/ldmadmin
   ```

6. In Catalog Administrator, click **Resource**, and then select **New > Resource**.

   The create resource view opens.

7. Create each Hive and HDFS scanner resource.

   Enter the property values you copied on each tab in Catalog Administrator.

Step 3. Restore the Enterprise Data Preparation Connections

Import the connection objects for the HDFS, Hive, and Hadoop connections used by the Enterprise Data Preparation Service from the .xml file to which you exported the connection objects.

Use the infacmd isp ImportDomainObjects command to import the connection object for each connection from the .xml file.

For more information about using the infacmd isp ImportDomainObjects command, see the *Informatica Command Reference*.

Run the following command:

```
infacmd isp ImportDomainObjects -dn <domain name> -un <user name> -pd <password>
-sdn <security domain> -fp <import file>
```

The following table describes the infacmd isp ImportDomainObjects command options and arguments you might specify to import the connection objects for the HDFS, Hive, and Hadoop connections:

Option	Argument	Description
-DomainName -dn	domain_name	Required. Name of the Informatica domain.
-UserName -un	user_name	Required if the domain uses Native or LDAP authentication. User name to connect to the domain. Optional if the domain uses Kerberos authentication. To run the command with single sign-on, do not set the user name. If you set the user name, the command runs without single sign-on.
-Password -pd	password	Required if you specify the user name.
-SecurityDomain -sdn	security_domain	Required if the domain uses LDAP authentication. Optional if the domain uses native authentication or Kerberos authentication. Name of the security domain to which the domain user belongs. If the domain uses native or LDAP authentication, the default is Native. If the domain uses Kerberos authentication, the default is the LDAP security domain created during installation.
-ImportFilePath -fp	import_file_path	Required. Path and file name of the XML file from which you import the objects.

The following example imports the connection objects for the HDFS, Hive, and Hadoop connections from the specified .xml file:

```
infacmd isp ImportDomainObjects -dn InfaDomain -un Administrator -pd password
-fp "/backup/edl_connections.xml"
```

Step 4. Restore the Data Preparation Repository

Restore the Data Preparation repository, and then use the Administrator tool to configure the Interactive Data Preparation Service to use the repository schema.

1. Restore the MySQL database or Oracle database used as the Data Preparation repository.

   For more information, see the MySQL documentation or the Oracle documentation.
2. In the Administrator tool, select the **Services and Nodes** view.
3. In the Domain Navigator, select the Interactive Data Preparation Service.
4. In the **Properties** view, expand **Data Preparation Repository Options**.
5. Click the **Edit** icon.
6. Enter the properties required to connect to the Data Preparation repository.

The following table describes the repository properties:

Property	Description
Database Type	Type of database to use for the Data Preparation repository.
Host Name	Host name of the machine that hosts the Data Preparation repository database. This property appears only if your Database Type is MySQL.
Database Port Number	Port number for the database. This property appears only if the Database Type is MySQL.
Connection String	Connection used to access the Oracle database. Use the following connection string: `jdbc:informatica:oracle://<host name>:<port>;ServiceName=<database name>` This field appears if you select Oracle as your Database Type.
Secure JDBC Parameters	If the database is secured, information such as TrustStore and TrustStorePassword can be included in this field. It is saved in an encrypted format. Parameters usually configured include the following: *EncryptionMethod=<encryption method>;HostNameInCertificate=<hostname>;TrustStore=<truststore file with its location>;TrustStorePassword=<truststore password>;KeyStore==<keystore file with its location>;KeyStorePassword=<keystore password>;ValidateServerCertificate=<true /false>* This field appears if you select Oracle as your Database Type.
Database User Name	Database user account to use to connect to the Data Preparation repository.
Database User Password	Password for the Data Preparation repository database user account.
Modify Database User Password	Select this checkbox to modify the database user password.
Schema Name	Schema or database name of the Data Preparation repository database. This field appears if you select MySQL as your Database Type.

Step 5. Restore the Interactive Data Preparation Service Durable Storage

Import the Interactive Data Preparation Service durable storage metadata into a durable storage location in the HDFS. After you import the durable storage metadata, use the Administrator tool to configure the Interactive Data Preparation Service to use the durable storage location.

You can use the Hadoop DistCp tool to import the storage metadata from the backup location to the HDFS location.

1. On HDFS, run the following command:

```
hadoop distcp <backup location> hdfs://<durable storage location>
```

The following table describes the DistCp command arguments:

Argument	Description
Durable Storage Location	The HDFS storage location to which to copy the durable storage metadata.
Backup Location	The path to the backup directory from which to copy the durable storage metadata.

2. In the Administrator tool, select the **Services and Nodes** view.
3. In the Domain Navigator, select the Interactive Data Preparation Service.
4. In the **Properties** view, expand **Data Preparation Storage Options**, and then click the **Edit** icon.
5. Enter the following properties:

Property	Description
Local Storage Location	Directory for data preparation file storage on the node where the Interactive Data Preparation Service runs.
HDFS Connection	Connection to the durable storage location in HDFS.
HDFS Storage Location	Durable storage location in HDFS.
Hadoop Distribution Directory	Location of the Hadoop distribution on HDFS data nodes.

Step 6. Recycle the Application Services

Start the application services used by Enterprise Data Preparation.

Start the application services in the following order:

- Model Repository Service
- Data Integration Service
- Content Management Service
- Interactive Data Preparation Service
- Enterprise Data Preparation Service

1. In the Administrator tool, select the **Services and Nodes** view.
2. In the Domain Navigator of the Administrator tool, select an application service.
3. Click the **Recycle the Service** button.
4. Click **OK**.

Step 7. Restore the Rules Metadata

If rules are configured for the Interactive Data Preparation Service, restore the rules metadata to the Model repository used by the Interactive Data Preparation Service.

Use the infacmd tools ImportObjects command to import the rules metadata, the dictionary content, and the NER and Classifier model files into a project in the Model repository associated with the Interactive Data Preparation Service.

The Model Repository Service that manages the Model repository used by the Interactive Data Preparation Service must be running when you run the command.

For more information about using the infacmd tools ImportObjects command, see the *Informatica Command Reference*.

Run the following command:

```
 infacmd tools ImportObjects -dn <domain name> -un <user name> -pd <password>
-sdn <security domain> -tn <target project name> -rs <Model Repository Service name>
-fp <import file> -sp <source project name> -tf <target folder name> -oo <options>
```

The following table describes the infacmd tools ImportObjects options and arguments:

Option	Argument	Description
-DomainName -dn	domain_name	Name of the Informatica domain.
-UserName -un	user_name	User name to connect to the domain. Optional if the domain uses Kerberos authentication.
-Password -pd	password	Password for the user name.
-SecurityDomain -sdn	security_domain	Required if the domain uses LDAP authentication. Optional if the domain uses native authentication or Kerberos authentication. Name of the security domain to which the domain user belongs. If the domain uses native or LDAP authentication, the default is Native. If the domain uses Kerberos authentication, the default is the LDAP security domain created during installation.
-TargetProject -tp	target_project	Name of the project in the Model repository into which you want to import the objects. The project must exist in the Model repository before you import the objects. The option is ignored if you use an import control file.
-RepositoryService - rs	service_name	Model Repository Service name.
-ImportFilePath -fp	file_path	Path and file name of the XML file from which to import the rules metadata. You can specify an absolute path or a relative path to the file name.
-SourceProject -sp	project_name	Source project name in the file to import.

Option	Argument	Description
-TargetFolder -tf	folder_name	Target folder within the project into which you want to import the rules objects. If you do not specify a target folder, the objects are imported into the target project. The folder must exist in the repository before you import the objects.
-OtherOptions -oo	options	Additional options to import the dictionary content and the NER and Classifier model files from the specified zip file. Enter options using the following format: `rtm:<option_name>=<value>,<option_name>=<value>` Required option names include: - codePage. Code page of the reference data. - refDataFile. Path and file name of the zip file from where you want to import the reference table data.

The following example imports rules metadata from a dq_rules_metadata.xml file into a folder named dq_rules within a project named NEW_PROJECT. The example also imports the dictionary content and the NER and Classifier models from a dq_rules_artifacts.zip file into the project.

```
infacmd tools importObjects -dn InfaDomain -un Administrator -pd password -sdn native
-rs MRS_dev3 -tp NEW_PROJECT -tf dq_rules -fp "/export/dq_rules_metadata.xml" -ow true
-oo "rtm:codePage=UTF-8,refDataFile="/output/dq_rules_artifacts.zip""
```

CHAPTER 11

Managing the Data Lakehouse

This chapter includes the following topics:

- Data Lakehouse Management Overview, 65
- Viewing Resources in the Data Lakehouse, 66
- Adding a Resource to the Data Lakehouse, 66
- Deleting a Resource From the Data Lakehouse, 67
- Managing Access to Data Lakehouse Resources, 67
- Managing Connections in a Data Lakehouse, 68
- Managing Advanced Properties in a Data Lakehouse, 68

Data Lakehouse Management Overview

You can use the Enterprise Data Preparation application to manage the Enterprise Data Catalog resources that represent data sources from which scanners extract metadata for use in the data lake, and data lakehouse targets to which users upload and publish data assets.

You can add resources created in Enterprise Data Catalog to the data lakehouse. You must add at least one Hive resource, which is required to perform export, import, data preparation and publish operations in the Enterprise Data Preparation application, to the data lakehouse. Contact your Enterprise Data Catalog administrator for assistance with creating new resources in Enterprise Data Catalog.

You can also delete a resource from the data lakehouse. Deleting a resource might impact Enterprise Data Preparation operations, including export, import, data preparation and publish operations.

You can manage access to the data lakehouse by assigning permissions on the Enterprise Data Catalog resources that represent Hive schemas and HDFS locations in the data lakehouse to Enterprise Data Preparation users and groups of users. When you grant permissions on specific schemas or locations to a user or group of users, the application displays only those schemas and locations to which a user has permissions when the user performs an import, publish, or upload operation.

Viewing Resources in the Data Lakehouse

Use the Enterprise Data Preparation application to view Enterprise Data Catalog resources available in the data lakehouse.

The Lakehouse Management page in the Enterprise Data Preparation application provides the following details for each resource:

- The resource name.
- The resource type.
- When the resource was added to the data lakehouse.
- The name of the user who added the resource to the data lakehouse.

1. Select the **Settings** icon, and then select **App Configuration**.
2. Click the **Resource Management** tab.

   Available resources appear in the table below.

Adding a Resource to the Data Lakehouse

Use the Enterprise Data Preparation application to add Enterprise Data Catalog resources to the data lakehouse.

Enterprise Data Preparation supports the following data sources:

- Cloud Data Warehouse
  - Microsoft Azure Synapse
- Cloud Data Lake
  - Microsoft Azure Data Lake Storage
  - Amazon S3
- Cloud Database
  - Microsoft Azure SQL
- Data Warehouse
  - Teradata
- Database
  - Oracle
  - Microsoft SQL Server

You must add at least one Hive resource, which is required to perform export, import, data preparation and publish operations in the Enterprise Data Preparation application, to the data lakehouse.

You can add a cloud database and a data warehouse through JDBCv2 connector for data preview, preparation, and publication. You must have a license added to the Data Integration Service to use JDBCv2 connector. The cloud data warehouse, such Microsoft Azure SQL Data Warehouse, requires a native connector for data preview, preparation, and publication.

1. Select the **Settings** icon, and then select **App Configuration**.
2. Click the **Resource Management** tab.

3. Click **Add**.
4. Select a resource in the Add Resource panel, and then click **Add**.

Deleting a Resource From the Data Lakehouse

You can delete resources that are no longer used or needed from the data lakehouse.

Deleting a resource might impact Enterprise Data Preparation operations, including export, import, data preparation and publish operations. You can delete one resource at a time to prevent accidental deletions.

1. Select the **Settings** icon, and then select **App Configuration** > **Resource Management**.
2. Highlight the resource, and then click the **Delete** icon to the right.

Managing Access to Data Lakehouse Resources

You can manage access to the data lakehouse by assigning permissions on the Enterprise Data Catalog resources that represent Hive schemas and HDFS locations in the data lakehouse to Enterprise Data Preparation users and groups of users.

You can grant permissions to users or groups to perform the following operations on Hive schemas and HDFS locations in the data lakehouse:

- Delete assets from the data lakehouse.
- Download assets from the data lakehouse as comma-separated value (.csv) file or a Tableau Data Extract (.tde) files.
- Export publications from the data lakehouse to an external database.
- Import assets from external sources to a specific target in the data lakehouse.
- Publish or copy prepared assets to a specific target in the data lakehouse.
- Upload files to a specific target in the data lakehouse.

You can assign users or groups permissions to specific Hive schemas or HDFS locations represented by a resource, or to all Hive schemas or HDFS locations represented by the resource. When you grant permissions on specific schemas or locations to a user or group of users, the application displays only those schemas and locations to which a user has permissions when the user performs an import, publish, or upload operation.

Managing Data Lakehouse Permissions

You can assign, edit, or delete permissions for users or group to perform operations in the data lakehouse.

1. Select the **Settings** icon, and then select **App Configuration**.
2. Click the **Security Management** tab.
3. Click **Add**.
4. Start typing the user name of a user or group in the User/Group field, and then select the user or group to add.

To filter user names, click the filter icon, select **Add Field>User/Group Name**, and then enter the user's user name.

The application adds the unique identifier for the user or group, and the security domain to which the user or group belongs.

5. Select the resource on which to grant the user or group permissions.
6. Select the data lakehouse schema or folder associated with the resource on which to grant the user or group permissions.
7. Select **Grant** under each permission you want to grant to the user or group.
8. To edit a permission, hover over the user or group permission and click  and then save.
9. To delete a permission, hover over the user or group permission and click  .
10. Click **Save**.

Managing Connections in a Data Lakehouse

You can create a default mapping between a resource and connection in the **Connection Management**.

1. Select the **Settings** icon, and then select **App Configuration**.
2. Click the **Connection Management** tab.
3. Click **Add**.
4. Select a resource and a connection from the list.

   You can use a filter to list resources or connections based on the resource name, resource type, connection name, and connection type.
5. Click **Save**.

Managing Advanced Properties in a Data Lakehouse

You can configure the advanced properties for the Enterprise Data Preparation.

When you add or edit the advanced properties, you might need to log in again to reflect the changes.

Upload and Download Options

Configure the following options:

Max File Size Upload	Maximum size of the files that the users upload. The default value is 1,024 MB.
Max Rows To Download	Maximum rows that are exported to a CSV file for an asset. You can specify a maximum of 2,000,000,000. Enter a value of -1 to export all rows.

Data Preparation Options

Configure the following options:

Default Data Prep Sample Size	The default number of sample rows to fetch for data preparation. You can specify a maximum number of 1,000,000 rows and a minimum of 1 rows.
Max Data Prep Sample Size	Maximum number of rows to fetch for data preparation. You can specify a maximum number of 1,000,000 rows and a minimum of 1 rows.
Execution Engine for Hive Sampling	Engine for Hive sampling such as Hadoop Cluster, MR, Spark, or Tez. The default is Hadoop Cluster.

Visualization Options

Configure the following options:

Zeppelin URL	The URL to access the Zeppelin framework. The URL must be in `http[s]://<Zeppelin host name>:<port>`

Catalog Options

Configure the following options:

Max Recommendations to Display	Maximum number of recommended data assets to display on a **Projects** > **Add Worksheet** page. You can specify a maximum of 50 recommendations. A value of 0 means no recommendations will be displayed.
Enable Business Title	Displays the business glossary terms, display names, and Axon terms in the data preview or preparation of the assets.

Hadoop Options

Configure the following options:

Hive Staging Connection	Hive connection data preview and preparation of complex files.
Hive Table Storage Format	Stores the data in the format such as Hadoop Cluster, ORC, or Parquet. The default is Hadoop Cluster.

Help Options

Configure the following options:

Documentation	Select online or offline help for reference.
Enable Help Video	Enables or disables the help video on the home page.

CHAPTER 12

Schedule Export, Import and Publish Activities

This chapter includes the following topics:

- Scheduling Overview, 70
- Scheduler Service User Privileges, 70

Scheduling Overview

When a user imports, exports, or publishes a data asset, the user can choose to complete the activity at that time, or can choose to schedule the activity. Scheduling an activity enables users to import, export or publish updated data assets to the data lake on a recurring basis.

If a user decides to schedule an activity, the user can create a new schedule, or can select an existing schedule. Other users can use schedules that the user creates, and the user can use schedules created by other users.

As an administrator, you can create, edit or delete schedules as needed. For information on creating, editing, or deleting schedules, see the *Informatica Enterprise Data Preparation User Guide*.

The Scheduler Service must be running to create or use schedules in Enterprise Data Preparation.

Scheduler Service User Privileges

Users must be assigned privileges in the Scheduler Service to view, create, edit, or delete schedules and scheduled activities.

To enable a user to view, create, edit, or delete schedules and scheduled activities, assign the user the following privileges in the Scheduler Service:

- Create Schedules
- Edit Schedules
- Delete Schedules
- View Schedules

To enable a user to view schedules and select a schedule to use for a scheduled activity, assign the user the following privileges in the Scheduler Service:

- Edit Schedules
- View Schedules

CHAPTER 13

Interactive Data Preparation Service

This chapter includes the following topics:

- Interactive Data Preparation Service Overview, 72
- Before You Create the Interactive Data Preparation Service, 73
- Creating and Managing the Interactive Data Preparation Service, 73
- Interactive Data Preparation Service Properties, 78
- Interactive Data Preparation Service Process Properties, 82
- Configuring Interactive Data Preparation Service on Grid for Scalability, 84

Interactive Data Preparation Service Overview

The Interactive Data Preparation Service is an application service that manages data preparation within the Enterprise Data Preparation application.

When an analyst prepares data in a project, the Interactive Data Preparation Service connects to the Data Preparation repository to store worksheet metadata. The service connects to the Hadoop cluster to read sample data or all data from the Hive table, depending on the size of the data. The service connects to the HDFS system in the Hadoop cluster to store the sample data being prepared in the worksheet.

The Interactive Data Preparation Service uses an Oracle database, a MySQL database, or a MariaDB database for the data preparation repository. You must configure a local storage location for data preparation file storage on the node on which the service runs.

You can create the Interactive Data Preparation Service when you install Enterprise Data Preparation, or you can use the Administrator tool to create the service after installation. Create the Interactive Data Preparation Service before you create the Enterprise Data Preparation Service. When you create the Enterprise Data Preparation Service, you must associate it with a Interactive Data Preparation Service.

Before You Create the Interactive Data Preparation Service

Before you create the Interactive Data Preparation Service, complete the prerequisite tasks for the service.

If you use Oracle as the database for the Data Preparation repository, perform the following tasks before you create the Interactive Data Preparation Service:

- Set up the Oracle server database version 12cR2 that the Interactive Data Preparation Service connects to. Ensure that the database is case insensitive.
- Set up the required user account for the Oracle database with create, drop, and alter permissions for tables and views.

If you use MySQL or MariaDB as the database for the Data Preparation repository, perform the following tasks before you create the Interactive Data Preparation Service:

- Set up the MySQL server database version 5.6.26 or above that the Interactive Data Preparation Service connects to. Ensure that the database is case insensitive. For MySQL version 5.6.26 and above, set lower_case_table_names=1 and for MySQL version 5.7 and above, set explicit_defaults_for_timestamp=1 in the my.cnf file.
- Set up the required user account for the MySQL database with create, drop, and alter permissions for tables and views.

If the domain is secure, you must secure the services that you create for use by Enterprise Data Preparation.

- The following services in the domain and the YARN application must share the same common truststore file:
  - Data Integration Service
  - Model Repository Service
  - Catalog Service
  - Interactive Data Preparation Service
  - Enterprise Data Preparation Service
- The Interactive Data Preparation Service and Enterprise Data Preparation Service must also share the same keystore file.
- You can use different keystore files for the Data Integration Service, Model Repository Service, and Catalog Service. If you use different keystore files, you must add certificates corresponding to each of the keystores into a common truststore file.
- If you have configured Enterprise Data Preparation with one primary node and one or more backup nodes, you must copy the truststore files to a common directory and specify the same directory path for all nodes assigned to Enterprise Data Preparation.

Creating and Managing the Interactive Data Preparation Service

Use the Administrator tool to create and manage the Interactive Data Preparation Service. When you change a service property, you must recycle the service or disable and then enable the service for the changes to take affect.

Create the Interactive Data Preparation Service

If you did not create the Interactive Data Preparation Service service during the console, or if you ran the silent installer, create the service through the Administrator tool.

Before you create the Interactive Data Preparation Service, verify that you have created and enabled the following services:

1. In the Administrator tool, click the **Manage** tab.
2. Click the **Services and Nodes** view.
3. In the Domain Navigator, select the domain.
4. Click **Actions** > **New** > **Interactive Data Preparation Service.**
5. Enter the following properties:

Property	Description
Name	Name of the Interactive Data Preparation Service. The name is not case sensitive and must be unique within the domain. It cannot exceed 128 characters or begin with @. It also cannot contain spaces or the following special characters: ` ~ % ^ * + = { } \ ; : ' " / ? . , < > | ! ( )] [
Description	Description of the Interactive Data Preparation Service. The description cannot exceed 765 characters.
Location	Location of the Interactive Data Preparation Service in the Informatica domain. You can create the service within a folder in the domain.
License	License object with the data lake option that allows the use of the Interactive Data Preparation Service.
Node Assignment	Type of node in the Informatica domain on which the Interactive Data Preparation Service runs. Select **Single Node** if a single service process runs on the node or **Primary and Backup Nodes** if a service process is enabled on each node for high availability. However, only a single process runs at any given time, and the other processes maintain standby status. The **Primary and Backup Nodes** option will be available for selection based on the license configuration. Select the **Grid** option to ensure horizontal scalability by using a grid with multiple Interactive Data Preparation Service nodes. Improved scalability supports high performance, interactive data preparation during increased data volumes and number of users. Each user is assigned a node in the grid using round-robin method to distribute the load across the nodes. Default is Single Node.
Node	Name of the node on which the Interactive Data Preparation Service runs.

6. Click **Next**.

7. Enter the following Data Preparation repository properties:

Property	Description
Database Type	Type of database to use for the Data Preparation repository.
Database User Name	Database user account to use to connect to the database.
Database User Password	Password for the database user account.
Host Name	Host name of the database machine.
Port Number	Port number for the database.
Schema Name	Schema or database name of the Data Preparation repository database.

Connection String	Connection string used to connect to the database.
	To connect to an Oracle database, format the string as follows:
	`jdbc:informatica:oracle://<database host name>:<port>;ServiceName=<service name>`
	To connect to a non-secure MySQL or MariaDB database, format the string as follows:
	`jdbc:mysql://<database host name>:<port>`
	The connection string is optional if you connect to a non-secure database.
	To connect to an SSL-enabled MySQL or MariaDB database, format the string as follows:
	`verifyServerCertificate=true&useSSL=true&requireSSL=true`

Secure JDBC Parameters	Secure JDBC parameters required to access a secure database.
	To connect to a secure Oracle database, format the string as follows:
	`EncryptionMethod=SSL;HostNameInCertificate=<secure database host name>;ValidateServerCertificate=true`
	To connect to as secure MySQL or MariaDB database, format the string as follows:
	`trustCertificateKeyStoreUrl=file://<truststore path/truststore file name>&trustCertificatekeyStorePassword=<truststore password>`

8. Click **Next**.
9. Enter the following storage properties:

Property	Description
Local Storage Location	Directory for data preparation file storage on the node where the service runs.
Durable Storage Type	Storage type for the data preparation file.
Durable Storage Connection	Connection for the data preparation file storage.

Property	Description
Durable Storage Location	Location for the data preparation file storage. If the connection to the local storage fails, the service recovers data preparation files from the location. If the Hadoop cluster uses Kerberos authentication, the impersonation user name must have read, write and execute permission on the HDFS storage location directory. The default location is: /datalake/dps_durable_storage.
Hadoop Authentication Mode	Security mode enabled for the Hadoop cluster for data preparation storage. If the Hadoop cluster uses Kerberos authentication, you must set the required Hadoop security properties for the cluster.

10. Click **Next**.
11. Enter the logging properties.
12. If you plan to use rules, you must associate the Interactive Data Preparation Service with the Model Repository Service that manages the Model repository that contains the rule objects and metadata. You must also associate a Data Integration Service that runs rules during data preparation with the Interactive Data Preparation Service.

    Enter the following properties required to enable rules:

Property	Description
Enable Rule Execution	Enables the execution of the validation rule objects.
Model Repository Service Name	Name of the Model Repository Service. The name is not case sensitive and must be unique within the domain. It cannot exceed 128 characters or begin with @. It also cannot contain spaces or the following special characters: ` ~ % ^ * + = { } \ ; : ' " / ? . , < > | ! ( )] [You cannot change the name of the service after you create it.
Model Repository Service User Name	User name to access the Model Repository Service.
Model Repository Service Password	Password to access the Model Repository Service.
Security Domain	Select the security domain to access the Model Repository Service.
Data Integration Service Name	Name of the Data Integration Service.

13. Click **Next**.

14. Enter the following HTTP configuration properties:

Property	Description
HTTP Port	Port number for the HTTP connection to the Interactive Data Preparation Service.
Enable Secure Communication	Use a secure connection to connect to the Interactive Data Preparation Service. If you enable secure communication, you must set all required HTTPS properties, including the keystore and truststore properties.
HTTPS Port	Port number for the HTTPS connection to the Interactive Data Preparation Service.
Keystore File	Path and the file name of keystore file that contains key and certificates required for HTTPS communication.
Keystore Password	Password for the keystore file.
Truststore File	Path and the file name of truststore file that contains authentication certificates for the HTTPS connection.
Truststore Password	Password for the truststore file.

15. Click **Next**.
16. Enter the following rules execution property:

Property	Description
Rules Server Port	Port used by the rules server managed by the Interactive Data Preparation Service. Set the value to an available port on the node where the service runs.

17. Click **Finish**.
18. Select the Interactive Data Preparation Service in the Domain Navigator, and then select **Actions** > **Create Repository** to create the repository contents.
19. Select **Actions** > **Enable Service** to enable the Interactive Data Preparation Service.

Enabling, Disabling, and Recycling the Interactive Data Preparation Service

You can enable, disable, and recycle the service from the Administrator tool.

1. In the Administrator tool, click the **Manage tab** > **Services and Nodes view**.
2. In the Domain Navigator, select the service.
3. On the **Actions** tab, select one of the following options:
   a. **Enable Service** to enable the service.
   b. **Disable Service** to disable the service.

      Choose the mode to disable the service in. Optionally, you can choose to specify whether the action was planned or unplanned, and enter comments about the action. If you complete these options, the information appears in the Events and Command History panels in the Domain view on the Manage tab.

c. **Recycle Service** to recycle the service.

Editing the Interactive Data Preparation Service

To edit the Interactive Data Preparation Service, select the service in the Domain Navigator and click the Properties view. You can change the properties while the service is running, but you must restart the service for the properties to take effect.

To edit the Interactive Data Preparation Service:

1. To edit specific properties, click the pencil icon in the selected properties area.
2. In the **Edit Properties** window, edit the required fields.
3. Click **OK**.
4. Click **Actions** > **Recycle Service**.
5. In the **Recycle Service** window, select the required options.
6. Click **OK** to restart the service.

Deleting the Interactive Data Preparation Service

Only users with ADMIN or WRITE permissions for the Interactive Data Preparation Service can delete the service.

To delete the Interactive Data Preparation Service:

1. On the **Manage** tab, select the **Services and Nodes** view.
2. In the Domain Navigator, select the Interactive Data Preparation Service.
3. Disable the Interactive Data Preparation Service by clicking **Actions** > **Disable Service** .
4. To delete the Interactive Data Preparation Service, click **Actions** > **Delete**.

Interactive Data Preparation Service Properties

To view the Interactive Data Preparation Service properties, select the service in the Domain Navigator and click the Properties View. You can edit the properties by clicking the pencil icon in the respective area, while the service is running, but you must restart the service for the properties to take effect. You can configure the following Interactive Data Preparation Service properties:

- General Properties
- Data Preparation Repository Options
- Data Preparation Storage Options
- Hive Security Options
- Hadoop Options
- Custom Properties

General Properties

General properties for the Interactive Data Preparation Service include the name, description, and the node in the Informatica domain that the service runs on.

To edit the general properties, click the pencil icon in the General Properties area. In the **Edit General Properties** window, edit the required fields.
The following table describes the general properties for the service:

Property	Description
Name	Name of the Interactive Data Preparation Service. The name is not case sensitive and must be unique within the domain. It cannot exceed 128 characters or begin with @. It also cannot contain spaces or the following special characters: ` ~ % ^ * + = { } \ ; : ' " / ? . , < > | ! ( )] [
Description	Description of the service. The description cannot exceed 765 characters.
License	License object with the data lake option that allows use of the service.
Node	Name of the node on which the service runs.
Node Assignment	Type of node in the Informatica domain on which the service runs. Select **Single Node** if a single service process runs on the node or **Primary and Backup Nodes** if a service process is enabled on each node for high availability. However, only a single process runs at any given time, and the other processes maintain standby status. The **Primary and Backup Nodes** option will be available for selection based on the license configuration. Default is Single Node.
Backup Nodes	If your license includes high availability, nodes on which the service can run if the primary node is unavailable.
Grid	If you selected Grid for node assignment, select the grid that you want to use for the Interactive Data Preparation Service. A grid ensures horizontal scalability. Improved scalability supports high performance, interactive data preparation during increased data volumes and number of users. Each user is assigned a node in the grid using round-robin method to distribute the load across the nodes.

Data Preparation Repository Options

To edit the data preparation repository options, click the pencil icon in the Data Preparation Repository Options area. In the **Edit Data Preparation Repository Options** window, edit the required fields.

Oracle

1. To use an Oracle database for the Data Preparation repository, select **Oracle** as the database type.
2. Enter the connection properties for the database.

The following table describes the connection properties:

Property	Description
Database User Name	Database user account to use to connect to the Data Preparation repository.
Database User Password	Password for the Data Preparation repository database user account.
Connection String	JDBC connection string to connect to the database. Format the string as follows: `jdbc:informatica:oracle://<database host name>:<port>;ServiceName=<service name>`
Secure JDBC Parameters	List of secure database parameters to connect to the database. Format the parameters as follows: `EncryptionMethod=SSL;HostNameInCertificate=<secure database host name>;ValidateServerCertificate=true`

MySQL

1. To use a MySQL database or a MariaDB database for the Data Preparation repository, select **MySQL** as the Database Type.
2. Enter the connection properties for the database.
   The following table describes the connection properties:

Property	Description
Database User Name	Database user account to use to connect to the database.
Database User Password	Password for the Data Preparation repository database user account.
Database Host Name	Host name of the machine that hosts the database.
Database Port Number	Port number for the database.
Schema Name	Schema or database name of the Data Preparation repository database.

Connection String	Connection string to connect to the database.
	To connect to a non-secure database, format the string as follows:
	`jdbc:mysql://<database host name>:<port>`
	The connection string is optional if you connect to a non-secure database.
	To connect to an SSL-enabled database, format the string as follows:
	`verifyServerCertificate=true&useSSL=true&requireSSL=true`

Secure JDBC Parameters	String containing the path and file name for the database truststore file, and the truststore password. Format the string as follows: `trustCertificateKeyStoreUrl=file://<truststore path/truststore file name>&trustCertificatekeyStorePassword=<truststore password>`

Data Preparation Storage Options

Data preparation storage options enables you to specify the local storage and HDFS location for data persistence.

To edit the data preparation storage options, click the pencil icon in the Data Preparation Storage Options area. In the **Edit Data Preparation Storage Options** window, edit the required fields.

The following table describes the data preparation storage options:

Property	Description
Local Storage Location	Directory for data preparation file storage on the node where the service runs.
Durable Storage Type	Storage type for the data preparation file.
Durable Storage Connection	Connection for the data preparation file storage.
Durable Storage Location	Location for the data preparation file storage. If the connection to the local storage fails, the service recovers data preparation files from the location. If the Hadoop cluster uses Kerberos authentication, the impersonation user name must have read, write and execute permission on the HDFS storage location directory. The default location is: /datalake/dps_durable_storage.
Hadoop Authentication Mode	Security mode of the Hadoop cluster for data preparation storage. If the Hadoop cluster uses Kerberos authentication, you must set the required Hadoop security properties for the cluster.

Logging Options

Logging options include properties for the severity level for service logs. Configure the Log Severity property to set the logging level.

To edit the data preparation logging options, click the pencil icon.
The following table describes the data preparation logging options:

Property	Description
Log Severity	Severity of messages to include in the logs. Select from one of the following values: - FATAL. Writes FATAL messages to the log. FATAL messages include nonrecoverable system failures that cause the service to shut down or become unavailable. - ERROR. Writes FATAL and ERROR code messages to the log. ERROR messages include connection failures, failures to save or retrieve metadata, service errors. - WARNING. Writes FATAL, WARNING, and ERROR messages to the log. WARNING errors include recoverable system failures or warnings. - INFO. Writes FATAL, INFO, WARNING, and ERROR messages to the log. INFO messages include system and service change messages. - TRACE. Write FATAL, TRACE, INFO, WARNING, and ERROR code messages to the log. TRACE messages log user request failures. - DEBUG. Write FATAL, DEBUG, TRACE, INFO, WARNING, and ERROR messages to the log. DEBUG messages are user request logs.
Log Directory	Location of the directory of log files.

Advanced Service Options

To edit the advanced service options, click the pencil icon.

The following table describes the advanced service options:

Property	Description
Enable Rule Execution	Set to true for enabling the rule execution.
Model Repository Service Name	Name of the Model Repository Service. The name is not case sensitive and must be unique within the domain. It cannot exceed 128 characters or begin with @. It also cannot contain spaces or the following special characters: ` ~ % ^ * + = { } \ ; : ' " / ? . , < > | ! ( )] [You cannot change the name of the service after you create it.
Model Repository Service User Name	User name to access the Model Repository Service.
Model Repository Service Password	Password to access the Model Repository Service.
Security Domain	Select the security domain to access the Model Repository Service.
Data Integration Service Name	Name of the Data Integration Service.

Custom Properties

Configure custom properties that are unique to specific environments. You might need to apply custom properties in special cases.

When you define a custom property, enter the property name and an initial value. Define custom properties only at the request of Informatica Global Customer Support.
To view the custom properties, select the service in the Domain Navigator and click the Properties view. You can change the properties while the service is running, but you must restart the service for the properties to take effect.

To edit the custom properties, click the pencil icon in the Custom Properties area. In the **Edit Custom Properties** window, edit the required fields.

Interactive Data Preparation Service Process Properties

A service process represents a service running on a node.

To configure properties for the Interactive Data Preparation Service processes, click the **Processes** view. Select the node to configure properties specific to that node.

You can edit service process properties such as the HTTP configuration, advanced options, and custom properties. You can change the properties while the Interactive Data Preparation Service process is running, but you must restart the process for the changed properties to take effect.

HTTP Configuration Options

The HTTP configuration options specify the HTTP or HTTPS port. The properties also specify the keystore file and truststore file to use when the Interactive Data Preparation Service process uses the HTTPS protocol.

To edit the HTTP configuration options, click the pencil icon in the HTTP Configuration Options area. In the **Edit HTTP Configuration Options** window, edit the required fields.

The following table describes the HTTP configuration options for a Interactive Data Preparation Service process:

Property	Description
HTTP Port	Port number for the HTTP connection to theInteractive Data Preparation Service.
Enable Secure Communication	Use a secure connection to the Interactive Data Preparation Service. If you enable secure communication, you must enter all required HTTPS options.
HTTPS Port	Port number for the HTTPS connection to the service.
Keystore File	Path and the file name of the keystore file that contains key and certificates required for the HTTPS communication.
Keystore Password	Password for the keystore file.
Modify Keystore Password	Select this checkbox if you want to modify the keystore password.
Truststore File	Path and the file name of the truststore file that contains authentication certificates for the HTTPS connection.
Truststore Password	Password for the truststore file.
Modify Truststore Password	Select this checkbox if you want to modify the truststore password.

Advanced Options

You can set the maximum heap size and Java Virtual Machine (JVM) options from the Advanced Options area.

To edit the advanced options, click the pencil icon in the Advanced Options area. In the **Edit Advanced Options** window, edit the required fields.

The following table describes the advanced options:

Property	Description
Maximum Heap Size	Maximum amount of RAM in megabytes to allocate to the Java Virtual Machine (JVM) that runs the service.
JVM Command Line Options	JVM command line options for the service processes.

Configuring Interactive Data Preparation Service on Grid for Scalability

The Interactive Data Preparation Service requires most memory and CPU resources for in-memory database to support high performance interactive data preparation. When too many users try to prepare data simultaneously, performance of the interactive preparation can decline. The administrator might need to upgrade the hardware to improve the performance levels. To support increased preparation data volumes, the administrator can achieve horizontal scaling by creating a Interactive Data Preparation Service Grid with multiple service nodes.

Each user is assigned a node in the grid using round-robin method to distribute the load across the nodes. Homogeneous combinations of nodes are allowed. You can combine nodes with the same operating system, same CPU, same memory, and security setup. This allows for seamless restoration of data after node failures, enabling the Interactive Data Preparation Service to be highly available.

1. Install the Enterprise Data Preparation binaries on every node that is part of the grid.
2. Select Grid while configuring the Interactive Data Preparation Service.
3. Ensure that all of the folder locations mentioned in the configuration are present in all the nodes.

You can add or remove nodes dynamically from a grid. When a node is added into an active grid, the Interactive Data Preparation Service process does not start automatically. The Enterprise Data Preparation administrator must enable the process in the **Processes** tab of the Interactive Data Preparation Service to start the process in the node.

Adding a New Node when the Interactive Data Preparation Service is Running

When you add a new node to the grid where the Interactive Data Preparation Service runs, the new node will be in the Disabled state.

1. Log in to the Administrator tool.
2. Click **Services**.
3. Select the Interactive Data Preparation Service from the list.
4. Click the **Processes** tab of the service.
5. Select the newly added node.
6. On the top right hand corner, click the **Enable** icon to start the process.

   A warning message appears.
7. Click **OK**.

Removing Interactive Data Preparation Service Nodes from the Grid

At least one node should be active for the Interactive Data Preparation Service to run.

When you shut down a node or a node goes down, it does not affect the Interactive Data Preparation Service as long as at least one node remains enabled in the grid. An active session will not be automatically recovered. An error will appear and the user must reconnect the session to continue. If all the nodes in a grid are removed or shut down, the Interactive Data Preparation Service is disabled.

Monitoring Interactive Data Preparation Service Node Status

You can troubleshoot by finding the state of the Interactive Data Preparation Service nodes in a grid at any given point in time.

To find the nodes of the service along with the state, connect to the Data Preparation repository and execute the following SQL query:

```
select node_id, node_ip, state, created_ts, node_port, isp_node_name from dp_physical
node;
```

The state column shows the current state of the node service. It can be in any of the following states:

- ACTIVE: The node is ready to take new user sessions.
- SUSPECTED_UNREACHABLE: The node cannot accept new sessions as peer-check operation is failing on that node. The node might not be completely down as the server may recover after a brief period of high load.

To find the user to node assignment, connect to the Data Preparation repository and execute the following SQL query:

```
select login_id, node_ip, a.node_id, isp_node_name from dp_physical_node a, dp_user u,
dp_user_to_node_map m where a.node_id = m.node_id and u.id = m.user_id;
```

CHAPTER 14

Enterprise Data Preparation Service

This chapter includes the following topics:

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- Before You Create the Enterprise Data Preparation Service, 87
- Creating and Managing the Enterprise Data Preparation Service, 88
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Enterprise Data Preparation Service Overview

Enterprise Data Preparation requires the Enterprise Data Preparation Service to complete operations. The Enterprise Data Preparation Service is an application service that runs the Enterprise Data Preparation application in the Informatica domain.

The Enterprise Data Preparation application allows data analysts to create data preparation projects. Each step of the data preparation project is stored in a recipe that is translated into a mapping for execution on the Informatica platform.

When an analyst uploads data, the Enterprise Data Preparation Service connects to the HDFS system in the Hadoop cluster to temporarily stage the data. When an analyst previews data, the Enterprise Data Preparation Service connects to the Hadoop cluster to read from the Hive table.

You can create the Enterprise Data Preparation Service when you install Enterprise Data Preparation, or you can use the Administrator tool to create the service after installation.

Before You Create the Enterprise Data Preparation Service

Before you create the Enterprise Data Preparation Service, complete the prerequisite tasks for the service.

Perform the following tasks before you create the Enterprise Data Preparation Service:

- Verify that the Informatica domain has the following services which must be associated with the Enterprise Data Preparation Service:
  - Data Integration Service
  - Model Repository Service
  - Catalog Service
  - Content Management Service must be configured if you want to use the data domain discovery feature. Ensure that the database client is installed on all nodes of the Hadoop cluster with the correct settings in the hadoopEnv.properties file.
  - Interactive Data Preparation Service

    Note: An Enterprise Data Preparation Service and a Interactive Data Preparation Service must have a one to one association. Do not associate one Enterprise Data Preparation Service with multiple Interactive Data Preparation Service instances, or one Interactive Data Preparation Service with multiple Enterprise Data Preparation Service instances.
- If you use HTTPS to connect to the Enterprise Data Preparation Service, verify the location and password of the keystore and truststore files.
  If the domain is secure, you must secure each Enterprise Data Preparation Service that you create in Enterprise Data Preparation. The Enterprise Data Preparation Service instances must use the same keystore and truststore files that the domain uses. If you use separate security certificates, you must add the security certificates for the Enterprise Data Preparation Service to the truststore and keystore file for the domain. You must use these keystore and truststore files for Enterprise Data Preparation. You must import the certificate file to the truststore location for the Interactive Data Preparation Service and Enterprise Data Preparation Service.

  You must also use the same truststore files for the following services:
  - Data Integration Service
  - Model Repository Service
  - Catalog Service
  - Interactive Data Preparation Service
  - Enterprise Data Preparation Service

If the domain is secure, you must secure the services that you create in Enterprise Data Preparation.

- The following services in the domain and the YARN application must share the same common truststore file:

  - Data Integration Service
  - Catalog Service
  - Interactive Data Preparation Service
  - Enterprise Data Preparation Service

  Note: You can use different keystore files for the Data Integration Service, Model Repository Service, and Catalog Service. If you use different keystore files, you must add certificates corresponding to each of the keystores into a common truststore file.

- The Interactive Data Preparation Service and Enterprise Data Preparation Service must also share the same keystore file.
- If you have configured Enterprise Data Preparation with one primary node and one or more backup nodes, you must copy the truststore files to a common directory and specify the same directory path for all nodes assigned to Enterprise Data Preparation.

Creating and Managing the Enterprise Data Preparation Service

Use the Administrator tool to create and manage the Enterprise Data Preparation Service. When you change a service property, you must recycle the service or disable and then enable the service for the changes to take affect.

Create the Enterprise Data Preparation Service

If you did not create the Enterprise Data Preparation Service service during the console, or if you ran the silent installer, create the service through the Administrator tool.

Before you create the Enterprise Data Preparation Service, verify that you have created and enabled the following services:

Catalog Service

Interactive Data Preparation Service

1. In the Administrator tool, click the **Manage** tab.
2. Click the **Services and Nodes** view.
3. In the Domain Navigator, select the domain.
4. Click **Actions** > **New** > **Enterprise Data Preparation Service.**
5. Enter the following properties:

Property	Description
Name	Name of the Enterprise Data Preparation Service. The name is not case sensitive and must be unique within the domain. It cannot exceed 128 characters or begin with @. It also cannot contain spaces or the following special characters: ` ~ % ^ * + = { } \ ; : ' " / ? . , < > | ! ( )] [
Description	Description of theEnterprise Data Preparation Service. The description cannot exceed 765 characters.
Location	Location of the Enterprise Data Preparation Service in the Informatica domain. You can create the service within a folder in the domain.
License	License object that allows the use of the Enterprise Data Preparation Service.

Property	Description
Node Assignment	Type of node in the Informatica domain on which the Enterprise Data Preparation Service runs. Select **Single Node** if a single service process runs on the node or **Primary and Backup Nodes** if a service process is enabled on each node for high availability. However, only a single process runs at any given time, and the other processes maintain standby status. The **Primary and Backup Nodes** option is available based on the license configuration. Default is Single Node.
Node	Name of the node on which theEnterprise Data Preparation Service runs.

6. Click **Next**.
7. Enter the following properties for the Model Repository Service:

Property	Description
Model Repository Service	Name of the Model Repository Service associated with the Enterprise Data Preparation Service.
Model Repository Service User Name	User account to use to log in to the Model Repository Service.
Model Repository Service User Password	Password for the Model Repository Service user account.

8. Click **Next**.
9. Enter the following properties for the Interactive Data Preparation Service, Data Integration Service, and Catalog Service:

Property	Description
Interactive Data Preparation Service	Name of the Interactive Data Preparation Service associated with the Enterprise Data Preparation Service.
Data Integration Service	Name of the Data Integration Service associated with the Enterprise Data Preparation Service.
Catalog Service	Name of the Catalog Service associated with the Enterprise Data Preparation Service.
Catalog Service User Name	User account to use to log in to the Catalog Service.
Catalog Service User Password	Password for the Catalog Service user account.

10. Click **Next**.

11. Enter the following execution properties:

Property	Description
Execution Engine	Engine to run the mappings.
Hadoop Connection	Hadoop connection for the data lakehouse.
HDFS Connection	HDFS connection for the Hadoop working directory.
Hadoop Working Directory	HDFS directory where the Enterprise Data Preparation Service copies temporary data and files necessary for the service to run. This directory must have permissions to enable users to upload data.
Hadoop Authentication Mode	Security mode of the Hadoop cluster for the data lake. If the Hadoop cluster uses Kerberos authentication, you must set the required Hadoop security properties for the cluster.
Local Working Directory	Local directory that contains the files downloaded from the Enterprise Data Preparation Service application, such as .csv or .tde files

12. Click **Next**.
13. Enter the following user event logging properties:

Property	Description
Log User Activity Events	Indicates whether the Enterprise Data Preparation Service logs user activity events.
Solr JVM Options	Solr JVM options required to connect to the specified JDBC port used to retrieve data from Zookeeper. Set to connect to Zookeeper from an external client.
Index Directory	Location of a shared NFS directory used by primary and secondary nodes in a multiple node installation.

14. Click **Next**.
15. Enter the logging properties.
16. Click **Next**.
17. Enter the following advanced properties:

Property	Description
Maximum Concurrent Upload/Download Activities	Maximum concurrent upload or download activities. You can specify a maximum of 2,000,000,000 activities to run concurrently. Enter a value of -1 (default) to run unbounded number of activities concurrently.

18. Click **Next**.

19. Enter the following properties:

Property	Description
HTTP Port	Port number for the HTTP connection to the Enterprise Data Preparation Service.
Enable Secure Communication	Use a secure connection to connect to the Enterprise Data Preparation Service. If you enable secure communication, you must enter all required HTTPS options.
HTTPS Port	Port number for the HTTPS connection to the Enterprise Data Preparation Service.
Keystore File	Path and the file name of keystore file that contains key and certificates required for the HTTPS connection.
Keystore Password	Password for the keystore file.
Truststore File	Path and the file name of the truststore file that contains authentication certificates for the HTTPS connection.
Truststore Password	Password for the truststore file.

20. Select **Enable Service** if you want to enable the service immediately after you create the service.

    If you want to enable the service at a later time, in the Domain Navigator, select the service and then select **Actions** > **Enable Service**.

21. Click **Finish**.

Enabling, Disabling and Recycling the Enterprise Data Preparation Service

You can enable, disable, and recycle the service from the Administrator tool.

1. In the Administrator tool, click the **Manage tab** > **Services and Nodes view**.
2. In the Domain Navigator, select the service.
3. On the **Actions** tab, select one of the following options:
   a. **Enable Service** to enable the service.
   b. **Disable Service** to disable the service.

      Choose the mode to disable the service in. Optionally, you can choose to specify whether the action was planned or unplanned, and enter comments about the action. If you complete these options, the information appears in the Events and Command History panels in the Domain view on the Manage tab.
   c. **Recycle Service** to recycle the service.

Editing the Enterprise Data Preparation Service

To edit the Enterprise Data Preparation Service, select the service in the Domain Navigator and click the Properties view. You can change the properties while the service is running, but you must restart the service for the properties to take effect.

To edit the Enterprise Data Preparation Service:

1. To edit specific properties, click the pencil icon in the selected properties area.

2. In the **Edit Properties** window, edit the required fields.
3. Click **OK**.
4. Click **Actions** > **Recycle Service**.
5. In the **Recycle Service** window, select the required options.
6. Click **OK** to restart the service.

Deleting the Enterprise Data Preparation Service

Only users with ADMIN or WRITE permissions for the Enterprise Data Preparation Service can delete the service.

To delete the Enterprise Data Preparation Service:

1. On the **Manage** tab, select the **Services and Nodes** view.
2. In the Domain Navigator, select the Enterprise Data Preparation Service.
3. Disable the Enterprise Data Preparation Service by clicking **Actions** > **Disable Service** .
4. To delete the Enterprise Data Preparation Service, click **Actions** > **Delete**.

Enterprise Data Preparation Service Properties

To view the Enterprise Data Preparation Service properties, select the service in the Domain Navigator and click the **Properties** view. You can edit the properties by clicking the pencil icon in the respective area, while the service is running, but you must restart the service for the properties to take effect. You can configure the following Enterprise Data Preparation Service properties:

- General Properties
- Repository Service Options
- Interactive Data Preparation Service Options
- Data Integration Service Options
- Catalog Options
- Execution Options
- Event Logging Options
- Logging Options
- Custom Options

General Properties

General properties for the Enterprise Data Preparation Service include the name, description, license, and the node in the Informatica domain that the Enterprise Data Preparation Service runs on.

To edit the general properties, click the pencil icon in the general properties area. In the **Edit General Properties** window, edit the required fields.

The following table describes the general properties for the service:

Property	Description
Name	Name of the Enterprise Data Preparation Service. The name is not case sensitive and must be unique within the domain. It cannot exceed 128 characters or begin with @. It also cannot contain spaces or the following special characters: ` ~ % ^ * + = { } \ ; : ' " / ? . , < > | ! ( )] [
Description	Description of the Enterprise Data Preparation Service. The description cannot exceed 765 characters.
License	License object with the data lake option that allows the use of the Enterprise Data Preparation Service.
Node Assignment	Type of node in the Informatica domain on which the Enterprise Data Preparation Service runs. Select **Single Node** if a single service process runs on the node or **Primary and Backup Nodes** if a service process is enabled on each node for high availability. However, only a single process runs at any given time, and the other processes maintain standby status. The **Primary and Backup Nodes** option will be available for selection based on the license configuration. Default is Single Node.
Node	Name of the node on which the Enterprise Data Preparation Service runs.
Backup Nodes	If your license includes high availability, nodes on which the service can run if the primary node is unavailable.

Model Repository Service Options

The Model Repository Service is an application service that manages the Model repository. When an analyst creates projects, the Model Repository Service connects to the Model repository to store the project metadata. When you create the Enterprise Data Preparation Service, you must associate it with a Model Repository Service using the Model Repository Service Options properties.

To edit the Model Repository Service options, click the pencil icon. In the **Edit Model Repository Service Options**window, edit the required fields.
The following table describes the Model Repository Service options:

Property	Description
Model Repository Service	Name of the Model Repository Service associated with the Enterprise Data Preparation Service.
Model Repository Service User Name	User account to use to log in to the Model Repository Service.
Model Repository Service Password	Password for the Model Repository Service user account.
Modify Repository Password	Select the checkbox to modify the Model Repository Service user password.
Security Domain	LDAP security domain for the Model repository user. The field appears when the domain contains an LDAP security domain.

Interactive Data Preparation Service Options

The Interactive Data Preparation Service is an application service that manages data preparation within the Enterprise Data Preparation application. When you create the Enterprise Data Preparation Service, you must associate it with a Interactive Data Preparation Service using the Interactive Data Preparation Service options.

To edit the Interactive Data Preparation Service options, click the pencil icon. In the **Edit Data Preparation Service Options**window, edit the required fields.
The following table describes the service options:

Property	Description
Interactive Data Preparation Service	Name of the Interactive Data Preparation Service associated with the Enterprise Data Preparation Service.

Data Integration Service Options

The Data Integration Service is an application service that performs data integration tasks for Enterprise Data Preparation. When you create the Enterprise Data Preparation Service, you must associate it with a Data Integration Service using the Data Integration Service options.

To edit the Data Integration Service options, click the pencil icon. In the **Edit Data Integration Service Options** window, edit the required fields.
The following table describes the Data Integration Service options:

Property	Description
Data Integration Service	Name of the Data Integration Service associated with the Enterprise Data Preparation Service.

Catalog Service Options

The catalog represents an indexed inventory of all the configured assets in an enterprise. You can find metadata and statistical information, such as profile statistics, data asset ratings, data domains, and data relationships, in the catalog. The catalog options will be based on the Catalog Service configuration you have set up when you installed Enterprise Data Catalog.

To edit the catalog service options, click the pencil icon in the Catalog Service Options area. In the **Edit Catalog Service Options** window, edit the required fields.

The following table describes the catalog service options:

Property	Description
Catalog Service	Name of the Catalog Service associated with the Enterprise Data Preparation Service.
Catalog Service User Name	User account to use to log in to the Catalog Service.
Catalog Service User Password	Password for the Catalog Service user account.

Property	Description
Modify Catalog Service User Password	Select this checkbox to modify the Catalog Service user password.
Security Domain	LDAP security domain for the Catalog Service user. The field appears when the domain contains an LDAP security domain.

Execution Options

Execution options include properties for the execution engine and the local system directory.

To edit the execution options, click the pencil icon in the Execution Options area. In the **Edit Execution Options** window, edit the required fields.

The following table describes the execution options:

Property	Description
Execution Engine	Engine for running the mappings.
Hadoop Connection	Hadoop connection for the data lakehouse.
HDFS Connection	HDFS connection for the Hadoop working directory.
Hadoop Working Directory	HDFS directory where the Enterprise Data Preparation Service copies temporary data and files necessary for the service to run. This directory must have permissions to enable users to upload data.
Hadoop Authentication Mode	Security mode of the Hadoop cluster for the data lakehouse. If the Hadoop cluster uses Kerberos authentication, you must set the required Hadoop security properties for the cluster.
Local System Directory	Local directory that contains the files downloaded from the Enterprise Data Preparation application, such as .csv or .tde files

Event Logging Options

Use the Event Logging Options area to configure user activity event logging options, optional Solr, and NFS directory properties.

To edit the event logging options, click the pencil icon. In the **Edit Event Logging Options** window, edit the required fields.
The following table describes the event logging options:

Property	Description
Log User Activity Events	Indicates whether the Enterprise Data Preparation Service logs user activity events for auditing.
JDBC Port	JDBC port to use to get audit events.

Property	Description
Solr JVM Options	Solr JVM options required to connect to the specified JDBC port used to retrieve data from Zookeeper. Set to connect to Zookeeper from an external client.
Index Directory	Location of a shared NFS directory used by primary and secondary nodes in a multiple node installation.

Logging Options

Logging options include properties for the severity level for service logs. Configure the Log Severity property to set the logging level.

To edit the logging options, click the pencil icon in the Logging Options area. In the **Edit Logging Options** window, edit the required fields.

The following table describes the logging options:

Property	Description
Log Severity	Severity of messages to include in the logs. Select from one of the following values: - FATAL. Writes FATAL messages to the log. FATAL messages include nonrecoverable system failures that cause the service to shut down or become unavailable. - ERROR. Writes FATAL and ERROR code messages to the log. ERROR messages include connection failures, failures to save or retrieve metadata, service errors. - WARNING. Writes FATAL, WARNING, and ERROR messages to the log. WARNING errors include recoverable system failures or warnings. - INFO. Writes FATAL, INFO, WARNING, and ERROR messages to the log. INFO messages include system and service change messages. - TRACE. Write FATAL, TRACE, INFO, WARNING, and ERROR code messages to the log. TRACE messages log user request failures. - DEBUG. Write FATAL, DEBUG, TRACE, INFO, WARNING, and ERROR messages to the log. DEBUG messages are user request logs.
Log Directory	Location of the directory of log files.

Custom Options

Configure custom properties that are unique to specific environments. You might need to apply custom properties in special cases.

When you define a custom property, enter the property name and an initial value. Define custom properties only at the request of Informatica Global Customer Support.
To view the custom options, select the service in the Domain Navigator and click the Properties view. You can change the properties while the service is running, but you must restart the service for the properties to take effect.

To edit the custom options, click the pencil icon in the Custom Options area. In the **Edit Custom Options** window, edit the required fields.

Enterprise Data Preparation Service Process Properties

A service process is the physical representation of a service running on a node. When the Enterprise Data Preparation Service runs on multiple nodes, a Enterprise Data Preparation Service process can run on each node with the service role. You can configure the service process properties differently for each node.

To configure properties for the Enterprise Data Preparation Service processes, click the **Processes** view. Select a node to configure properties specific to that node.

You can edit service process properties such as the HTTP port, advanced options, custom properties, and environment variables. You can change the properties while the Enterprise Data Preparation Service process is running, but you must restart the process for the changed properties to take effect.

HTTP Configuration Options

The HTTP configuration options specify the keystore and truststore file to use when the Enterprise Data Preparation Service uses the HTTPS protocol.

To edit the HTTP configuration options, click the pencil icon in the HTTP Configuration Options area. In the **Edit HTTP Configuration Options** window, edit the required fields. The following table describes the HTTP configuration options for a Enterprise Data Preparation Service process:

Property	Description
HTTP Port	Port number for the HTTP connection to the Enterprise Data Preparation Service.
Enable Secure Communication	Use a secure connection to connect to the Enterprise Data Preparation Service. If you enable secure communication, you must enter all required HTTPS options.
HTTPS Port	Port number for the HTTPS connection to the Enterprise Data Preparation Service.
Keystore File	Path and the file name of keystore file that contains key and certificates required for the HTTPS connection.
Keystore Password	Password for the keystore file.
Truststore File	Path and the file name of the truststore file that contains authentication certificates for the HTTPS connection.
Truststore Password	Password for the truststore file.

Advanced Options

You can set the maximum heap size and Java Virtual Machine (JVM) options from the Advanced Options area.

To edit the advanced options, click the pencil icon in the Advanced Options area. In the **Edit Advanced Options** window, edit the required fields.

The following table describes the advanced options:

Property	Description
Maximum Heap Size	Maximum amount of RAM in megabytes to allocate to the Java Virtual Machine (JVM) that runs the Enterprise Data Preparation Service.
JVM Command Line Options	JVM command line options for the Enterprise Data Preparation Service processes.
Maximum Concurrent Upload/Download Activities	Maximum concurrent upload or download activities. You can specify a maximum of 2,000,000,000 activities to run concurrently. Enter a value of -1 (default) to run unbounded number of activities concurrently.

Custom Options

Configure custom properties that are unique to specific environments. You might need to apply custom properties in special cases.

When you define a custom property, enter the property name and an initial value. Define custom properties only at the request of Informatica Global Customer Support.
To view the custom options, select the service in the Domain Navigator and click the Properties view. You can change the properties while the service is running, but you must restart the service for the properties to take effect.

To edit the custom options, click the pencil icon in the Custom Options area. In the **Edit Custom Options** window, edit the required fields.

Environment Variables

You can configure environment variables for the Enterprise Data Preparation Service process.

The following table describes the environment variables:

Property	Description
Environment variable	Enter a name and a value for the environment variable.

Apache Zeppelin Options

If Apache Zeppelin uses Spark 1.x version, you must specify the Spark version in an environment variable named sparkVersion in the Enterprise Data Preparation Service process properties.

You do not need to add the environment variable if Zeppelin uses a Spark 2.x version.

To add the environment variable, click the pencil icon in the Environment Variables area. The following table describes the sparkVersion environment variable:

Property	Description
sparkVersion	The Spark 1.x version used by Apache Zeppelin.

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